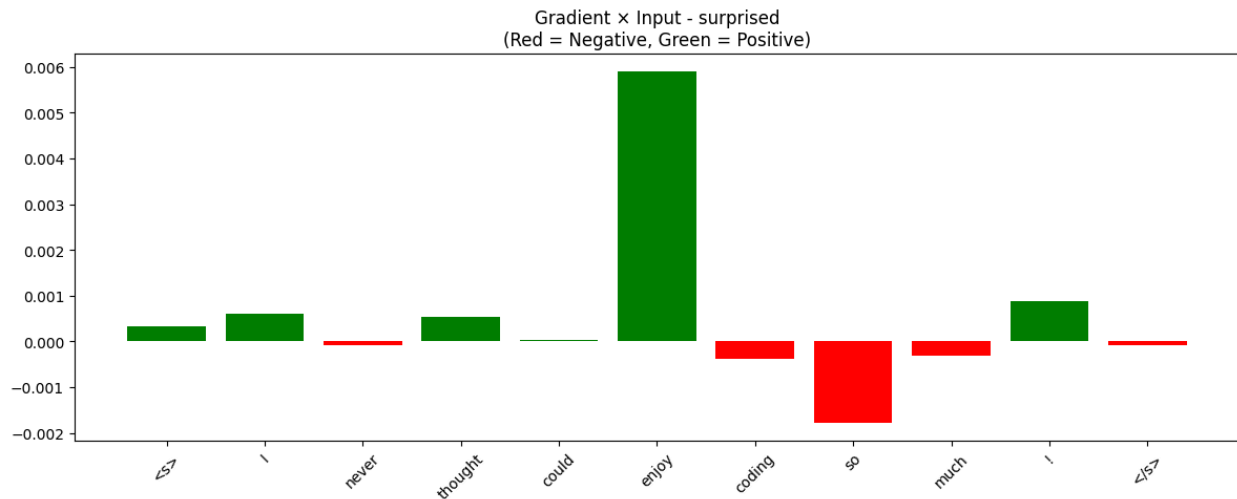
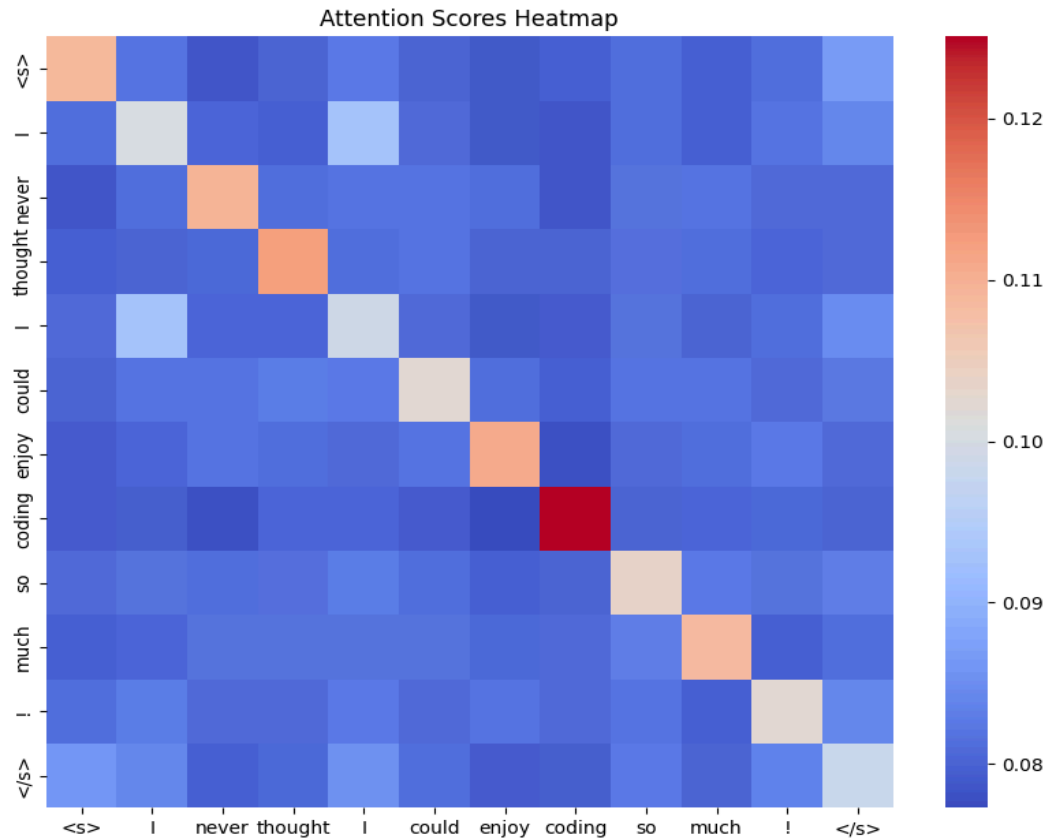


Sentence 1: "I never thought I could enjoy coding so much!"

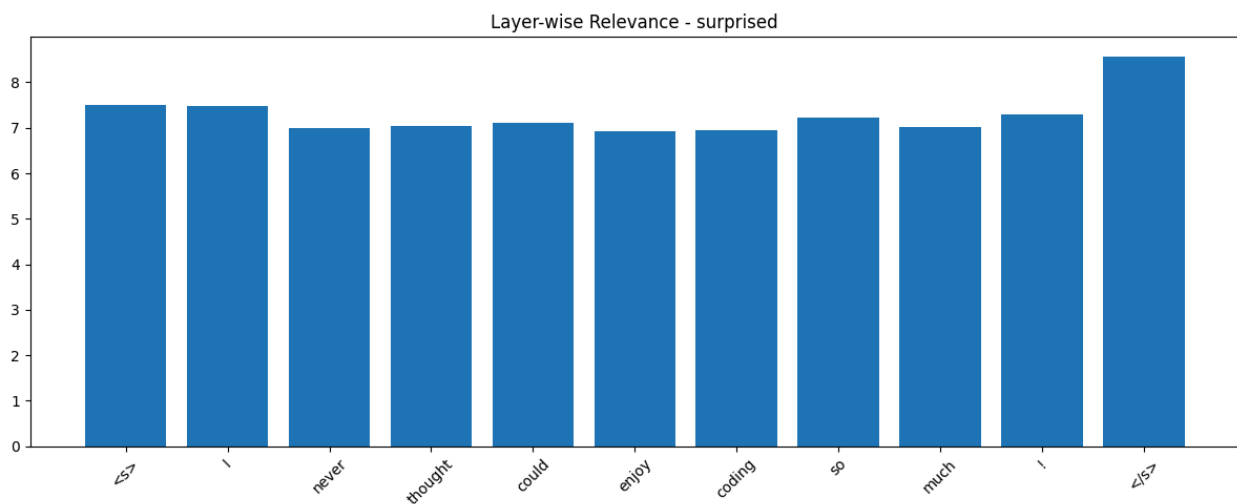
Predicted emotion: surprised



- It makes high relevance to "enjoy" as a good indicator of surprise, but rather for happiness than surprise, which aligns with human intuition. "Much" strengthens this positive contribution by emphasizing the degree of enjoyment, but its contribution is counterintuitive as it may seem to divert the model from the predicted class. "never" contributes negatively, which may be because it introduces a contrast between prior expectations and the current experience. "coding" is treated as a minor negative influence when I would have expected to be neutral or even slightly positive in the context of enjoyment. "so" has a disproportionate negative impact, even though I would actually think of it as a contributor to the overall emotion of surprise, which may mislead the model's classification. "!" is expected to have a positive contribution.

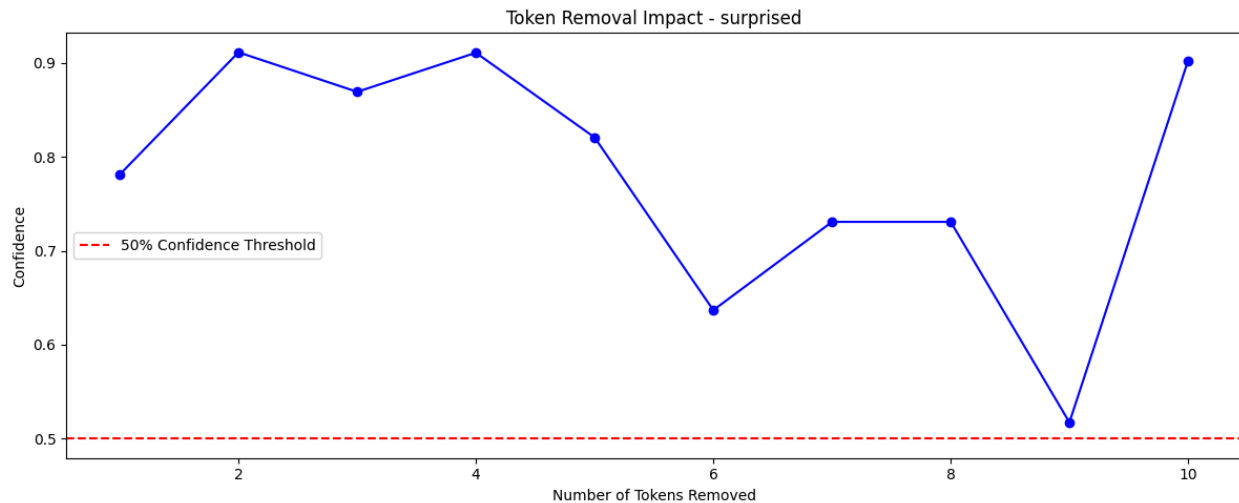


- The model heavily focuses on "coding" when processing it, treating it as important for emotion prediction, while mostly focusing on the self attention of the words.



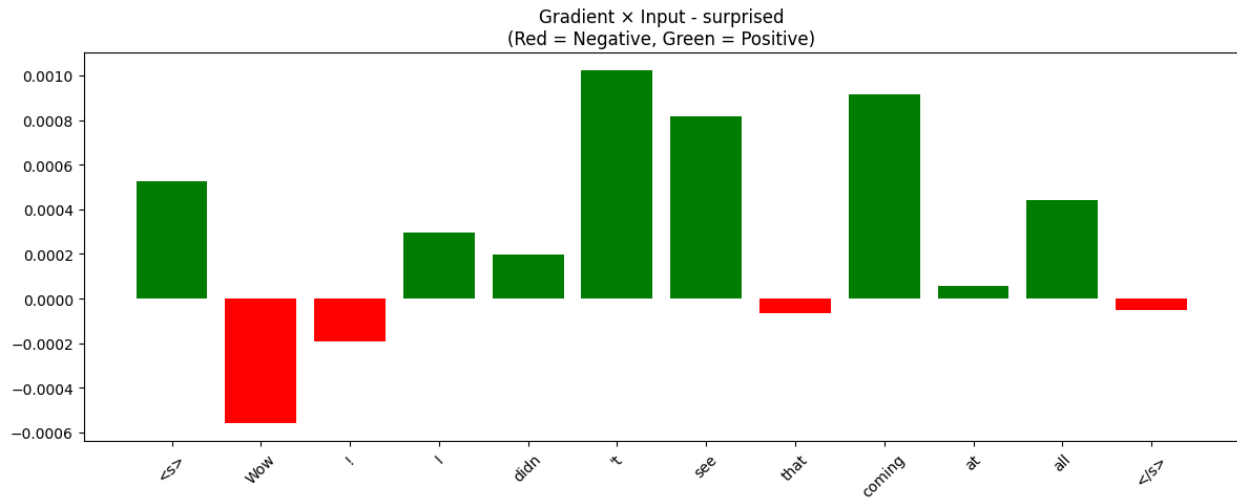
- The high relevance of "</s>" suggests that the model heavily relies on the end of the sentence to determine the emotion, capturing the transition from an initial expectation to

a realization. The relatively even distribution of relevance among words like "never," "could," "enjoy," "coding," "so," and "much" suggests that the model is considering the full sentence structure rather than disproportionately emphasizing any one word. While each of these words contributes to the overall meaning, their impact is more balanced compared to the Gradient $\times$ Input explanation.

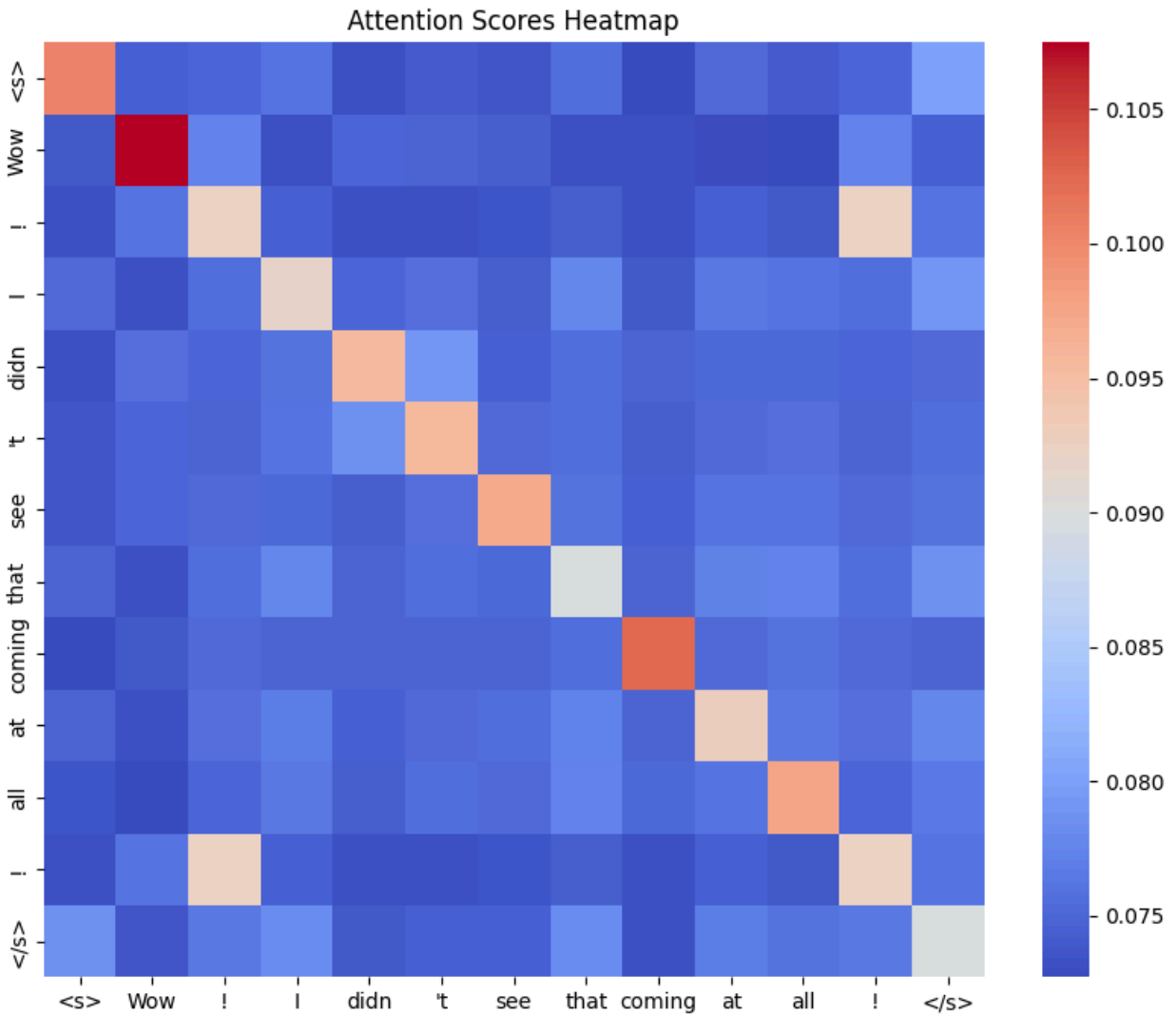


- If the model's confidence drops sharply after removing a few tokens, it means the model relies heavily on those specific words to make its prediction. If the confidence decreases gradually, the model spreads its attention across multiple tokens, using a broader range of context to determine the emotion. The significant drop in confidence when removing tokens in the middle-to-later stages (especially around $x = 4$ to 6 and $x = 9$) suggests that these tokens are essential for predicting "surprise." This aligns with earlier analyses that identified words like "enjoy" and "coding" as crucial emotional indicators. The sharp confidence drop at $x = 9$ indicates, this sensitivity suggests that the model's prediction relies on only a subset of critical words rather than distributing importance evenly across all words in the sentence.

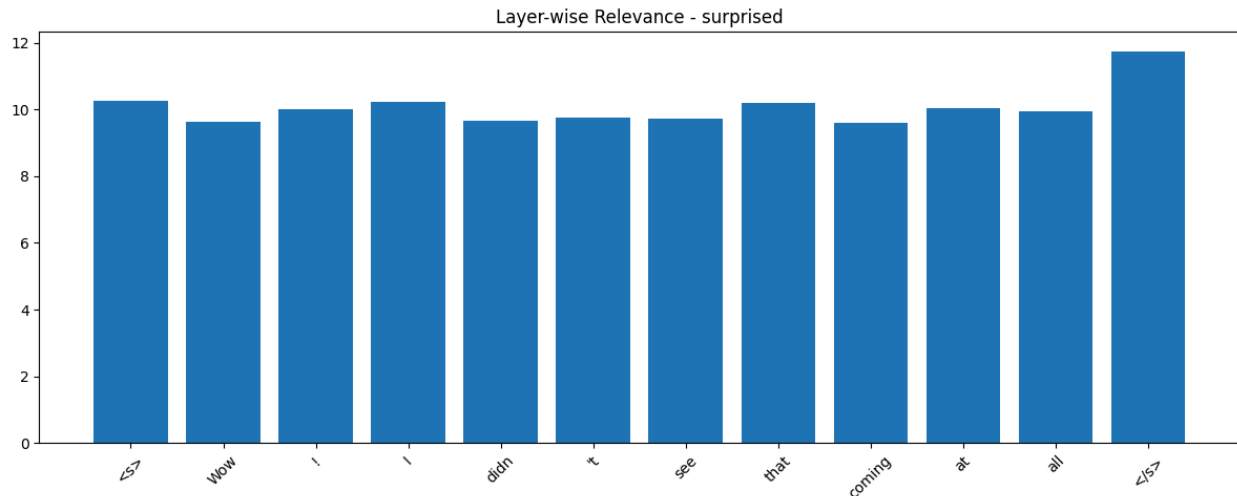
Sentence 2: "Wow! I did not see that coming at all!"
Predicted emotion: surprised



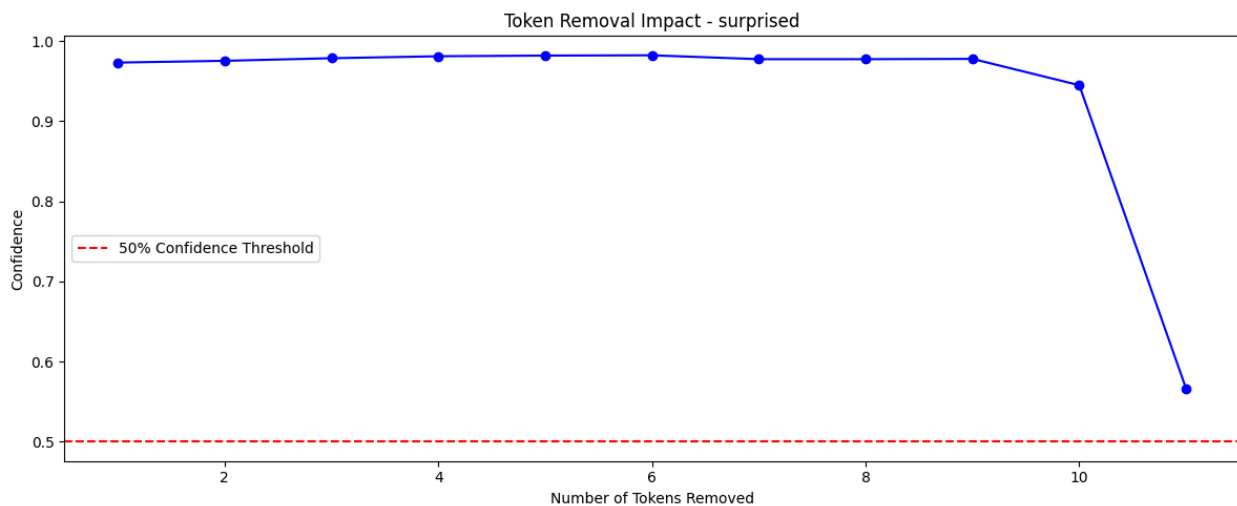
- In this case, the phrase "didn't see that coming at all" is identified as the strongest indicator of surprise, aligning well with human intuition. The words "didn't", "see", "coming", and "at all" contribute positively, amplifying the idea of an unexpected event. In everyday language, "Wow!" is often a strong indicator of surprise, rather than diminishing it. While "that" on its own may not carry strong emotional weight, its presence within the phrase should not detract from the overall sentiment.



- The model heavily focuses on the interjection "Wow", attending to itself, while also paying close attention to "coming", "didn't", and "all", recognizing them as key indicators of surprise. The model is relying on self-attention, where each token is primarily focused on itself.



- Gradient $\times$ Input previously assigned high relevance to "coming", treating it as a strong local indicator of surprise showing the unexpected events. However, in the LRP analysis, "coming" has a slightly lower relevance score, suggesting that the model now views it as more context-dependent rather than a standalone trigger of surprise. Both analyses evidence high relevance assigned to "</s>", the end-of-sentence token. Another difference lies in how the words "Wow", "didn't", "see", "that", and "all" are treated. In the Gradient $\times$ Input analysis, "Wow!" was unexpectedly assigned a negative contribution, implying that the model downplayed its role in conveying surprise. In the LRP analysis, however, these words now have more evenly distributed moderat.

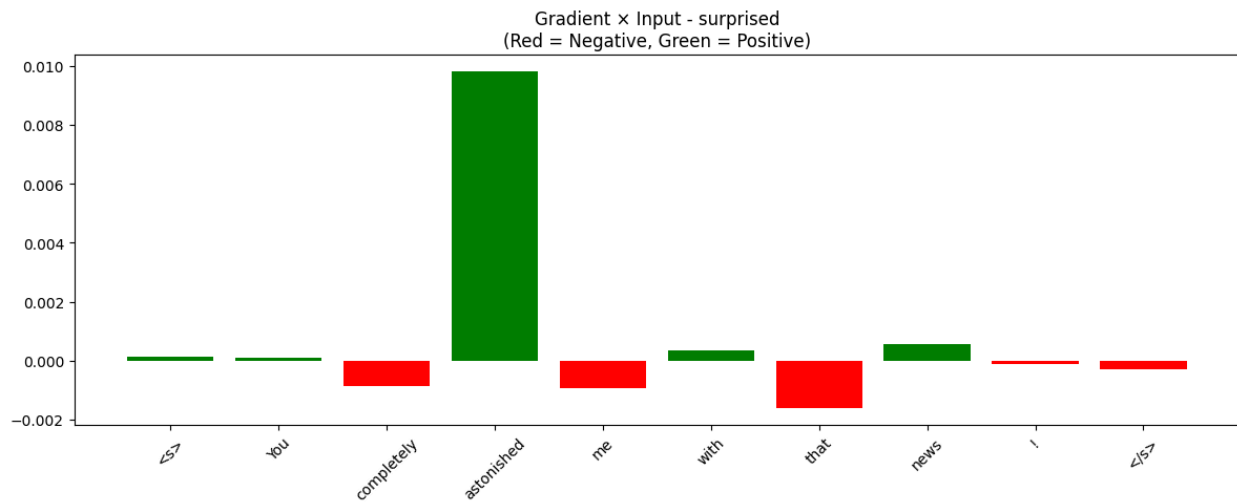


- The token removal shows that when no tokens are removed, the model's confidence is high at 0.97, indicating a strong prediction of "surprise." Removing the first 9 tokens does

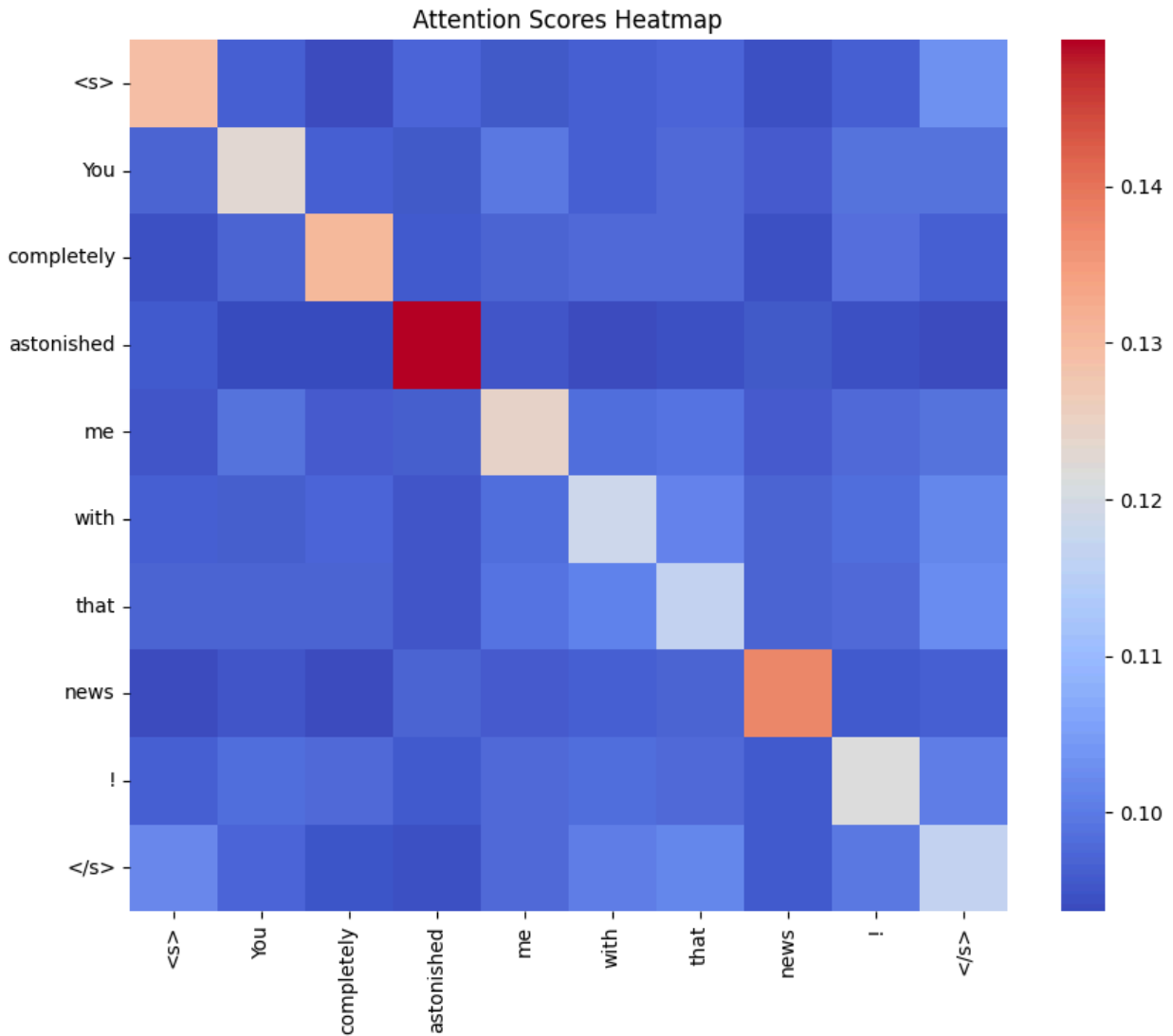
not impact the model's confidence showing that these tokens have little relevance to the prediction. However, the confidence drops significantly when the 9th to 10th and 11th tokens are removed—falling to 0.94 and 0.57, respectively. These tokens are crucial for predicting surprise, containing the strongest emotional signals like "didn't" or "coming." After removing these key tokens, the model's confidence drops to 0.57, but it still predicts "surprised," suggesting the remaining tokens provide enough context with less certainty. This indicates that the model is robust to the removal of less relevant tokens but heavily relies on a few key words to predict surprise.

Sentence 3: "You completely astonished me with that news!"

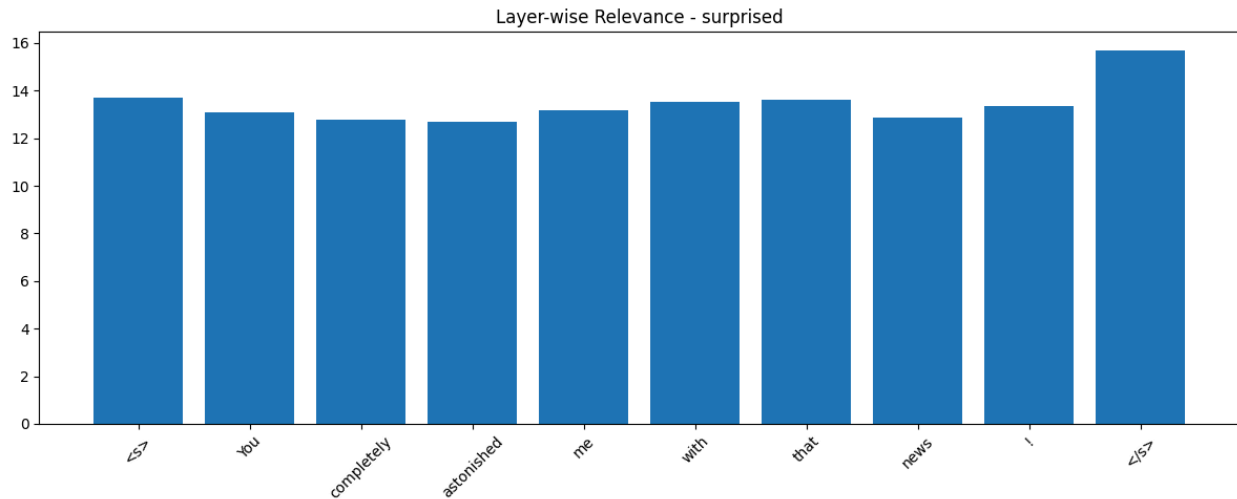
Predicted emotion: surprised



- The model correctly identifies "astonished" as key to predicting "surprised," aligning with my expectations. "<s>" slightly influences the model's surprise prediction, which is unexpected, as it doesn't carry emotion. "With" and "news" show slight positive contributions, linking surprise to its source, which I can understand. "completely," which seems to downplay surprise, "Me," "that," and "I" show negative effects, possibly reflecting a personal experience, which might reduce the emotional impact on the specific emotion of surprise. These words create contrasts that I wouldn't necessarily associate with diminished surprise.



- The model heavily focuses on the word "astonished," indicating its strong attention to this key emotion-indicating token. It also pays close attention to "news," recognizing it as surprising information. The word "completely" suggests the model is evaluating the intensity of surprise. The model is relying on self-attention, where each token is primarily focused on itself.



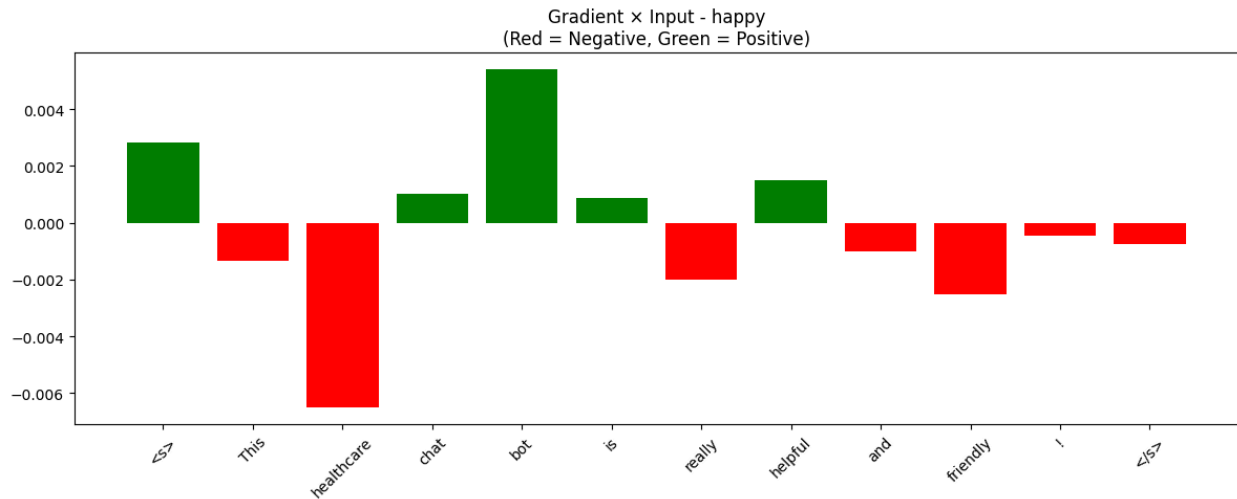
- The model now shows a more balanced focus on the sentence as a whole rather than overemphasizing emotionally charged tokens like "astonished." The model places moderate relevance on words like "completely," "news," "You," "me," "with," "that," and "I," showing that these contribute to the context of surprise. This is a change from the previous analysis, where the "completely" was given stronger negative weight, and "news" was recognized as a relevant part of the surprising information. The word "</s>" at the end of the sentence is assigned slightly higher relevance.



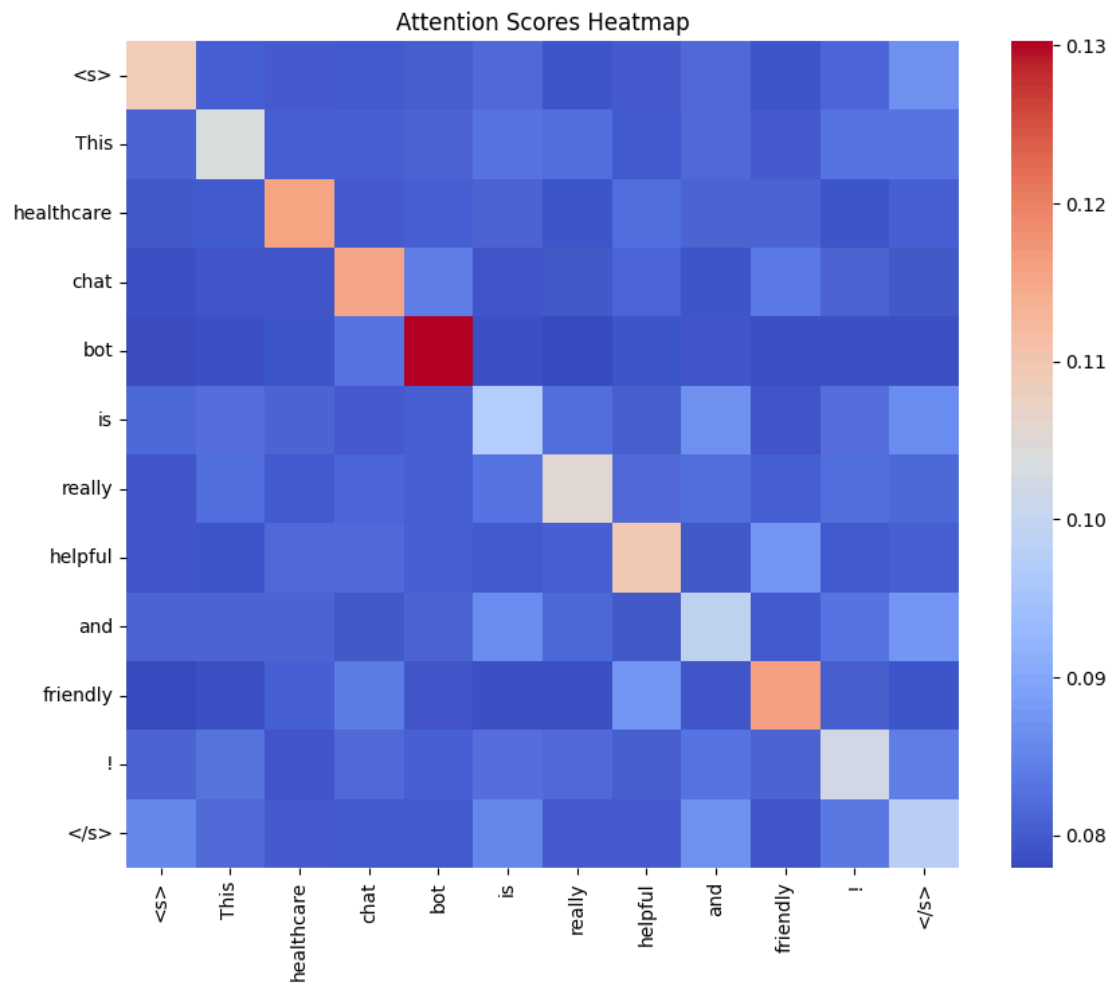
- The model's confidence is around 0.80 when no tokens are removed, indicating a good prediction of the emotion "surprised." When the second least relevant token is removed, the confidence drops sharply to around 0.40. This significant drop indicates that this token plays a crucial role in the model's prediction. As I continue removing the 3rd, 4th, 5th, and 6th least relevant tokens, the confidence fluctuates but remains generally below the initial confidence level. This indicates that the model distributes its reliance across

several tokens besides the second least relevant token. When the 7th and 8th least relevant tokens are removed, the model's confidence increases to around 0.92 at $x = 7$ and stays high at $x = 8$. This increase is counterintuitive. The fact that the model's confidence remains high even after removing the 8 least relevant tokens indicates that the remaining tokens provide enough information for the model to make a strong prediction of "surprise."

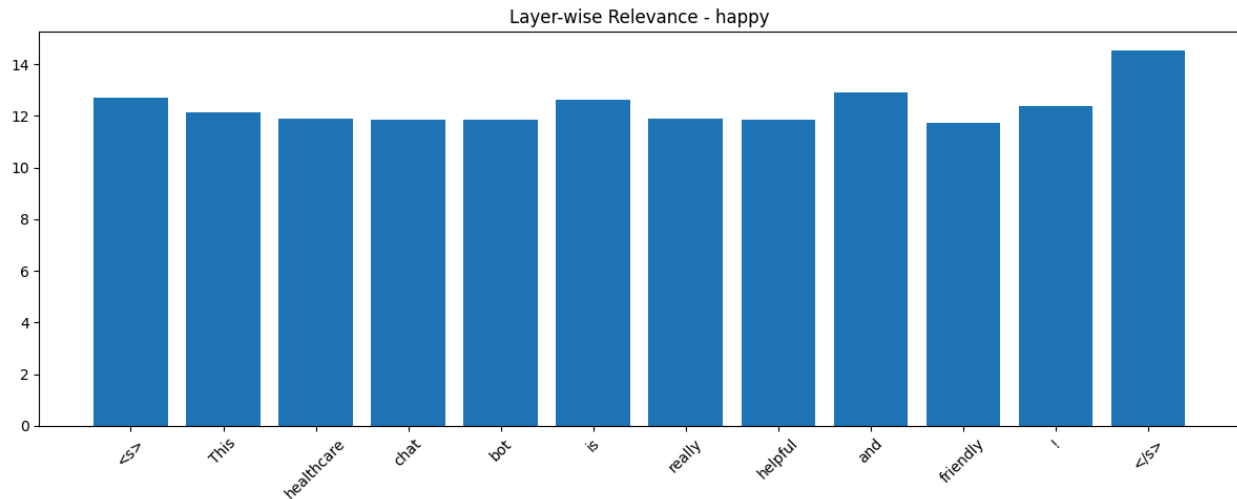
Sentence 4: "This healthcare chatbot is really helpful and friendly!"
Predicted emotion: happiness



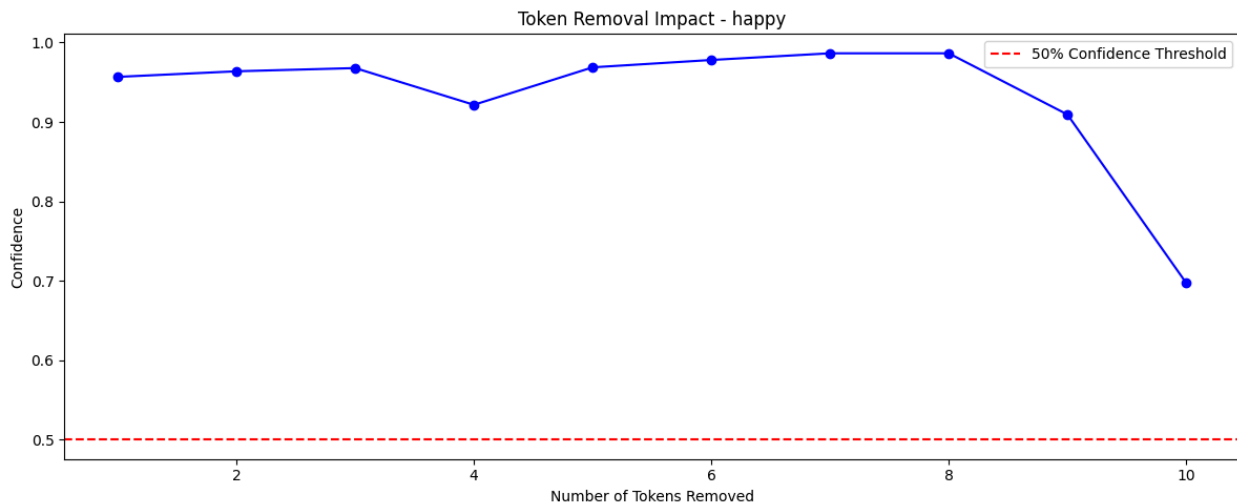
- The words "chat", "bot" receive a positive contribution, as expected, given its role in representing helpfulness in this context. "helpful" also contributes positively. The word "healthcare" is surprisingly negative. This could be due to its association with negative experiences in contexts like healthcare technology, showing bias which I did not expect. "And" and "Friendly" are unexpectedly negative and the first should be neutral. I would still assign the emotion of happiness.



- The model assigns high attention to the word "bot," indicating its importance as the subject of the sentence. Similarly, "helpful," "friendly," and "healthcare" each receive relatively high attention scores, suggesting they are key descriptors in the context.

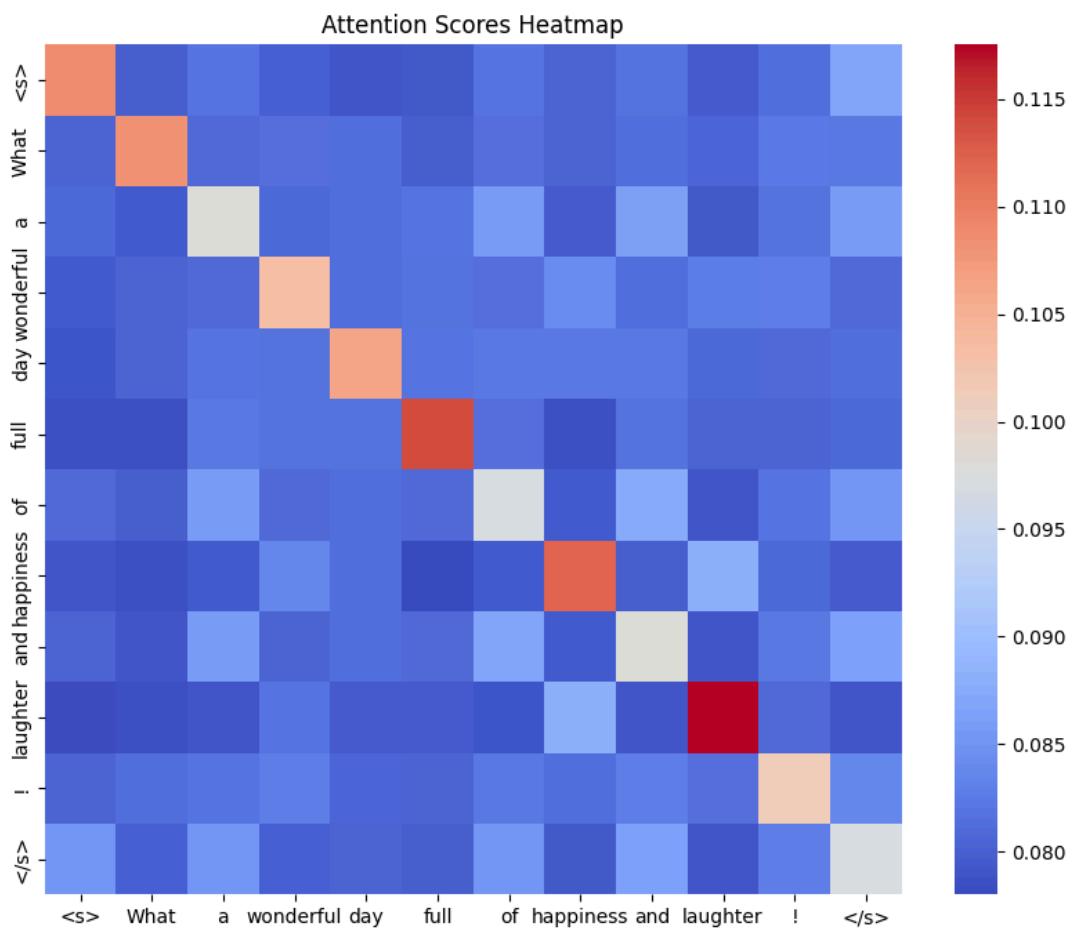
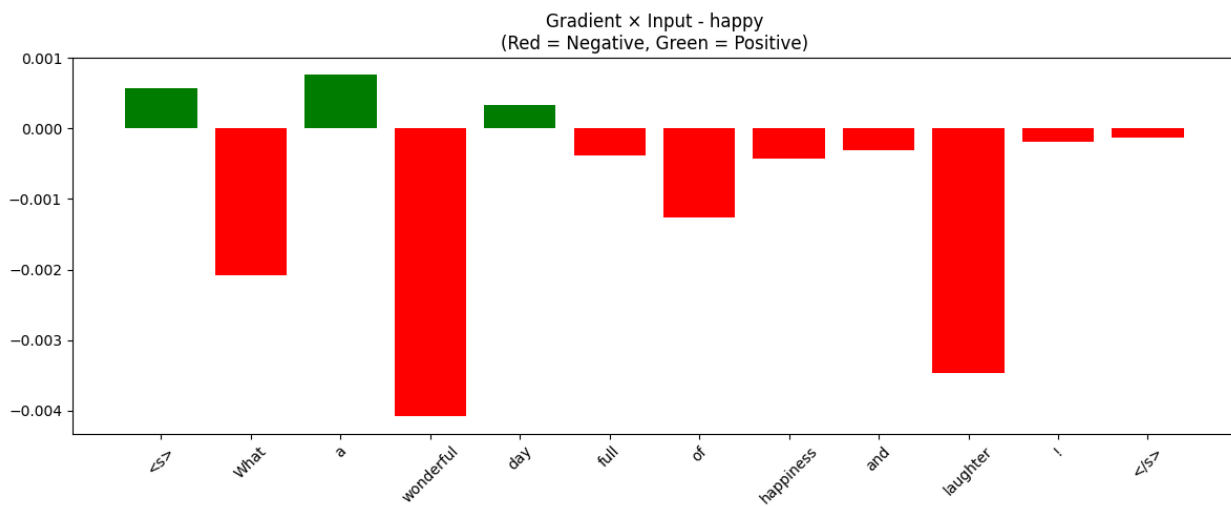


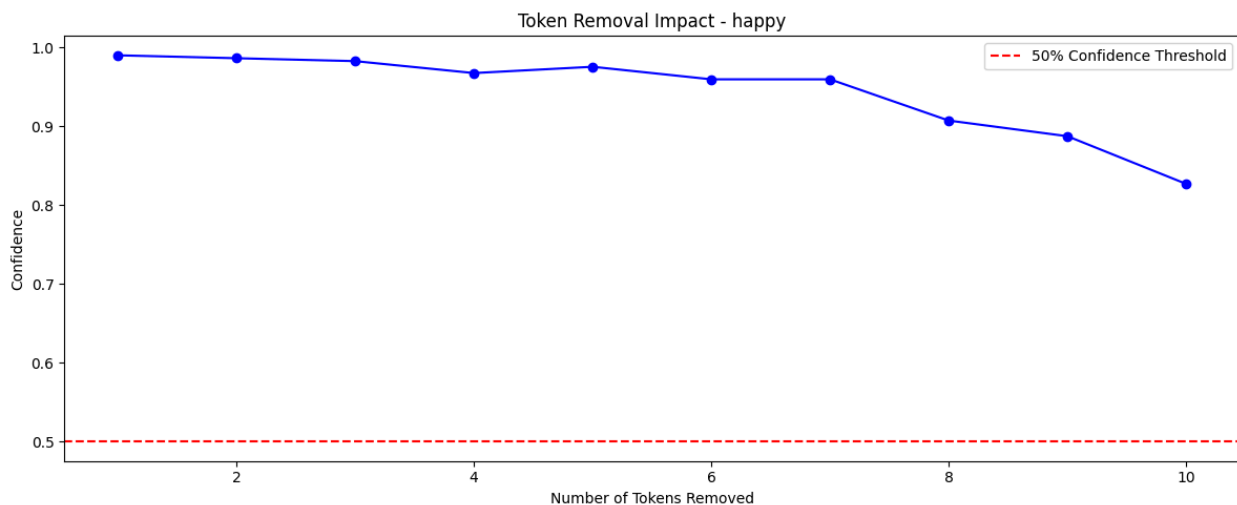
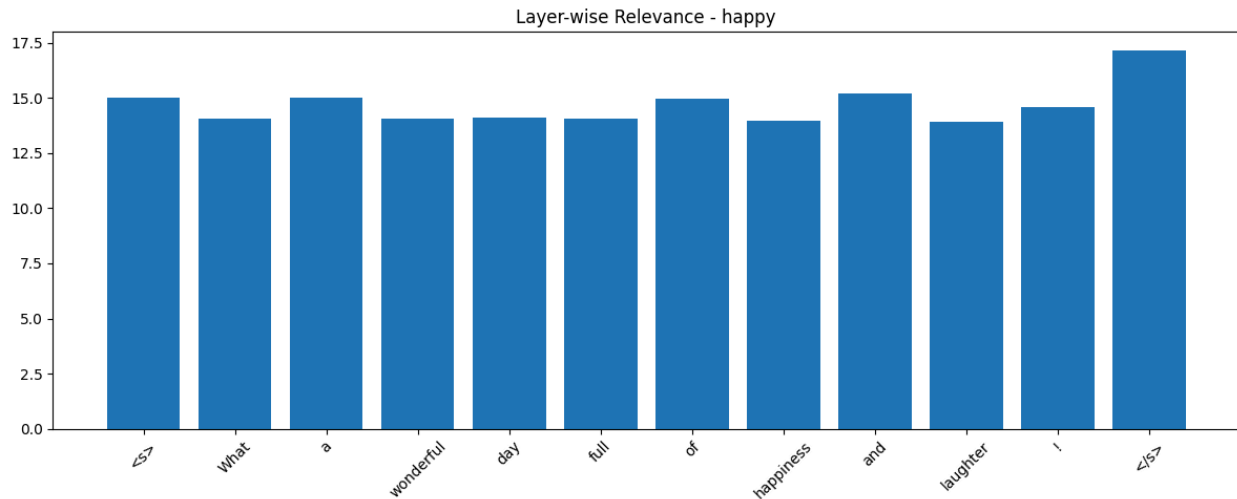
- In comparison to the first analysis, the model in the second explanation focuses more accurately on emotionally relevant tokens. The words "helpful" and "friendly" align with their strong contribution to positive sentiment. This shows improved focus on words directly showing this overall emotion. In contrast to the first explanation, where "healthcare" and "friendly" had surprising negative contributions, the second analysis treats these words as contextually important but less emotionally influential.



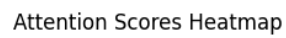
The analysis shows that the model's high initial confidence of 0.96 indicates a powerful, clear prediction of "happy." As tokens are removed ($x = 1$ to 8), the confidence stays above 0.9, suggesting the model uses a wide range of tokens to predict the sentiment, not relying on any single word. The positive descriptors like "helpful" and "friendly" help maintain a good prediction even with significant removal. However, the confidence drops sharply when 9 or 10 tokens are removed, indicating the model depends heavily on specific words like "helpful" and "friendly." Despite this drop, the confidence remains above 50%, showing the model still maintains great performance with less context.

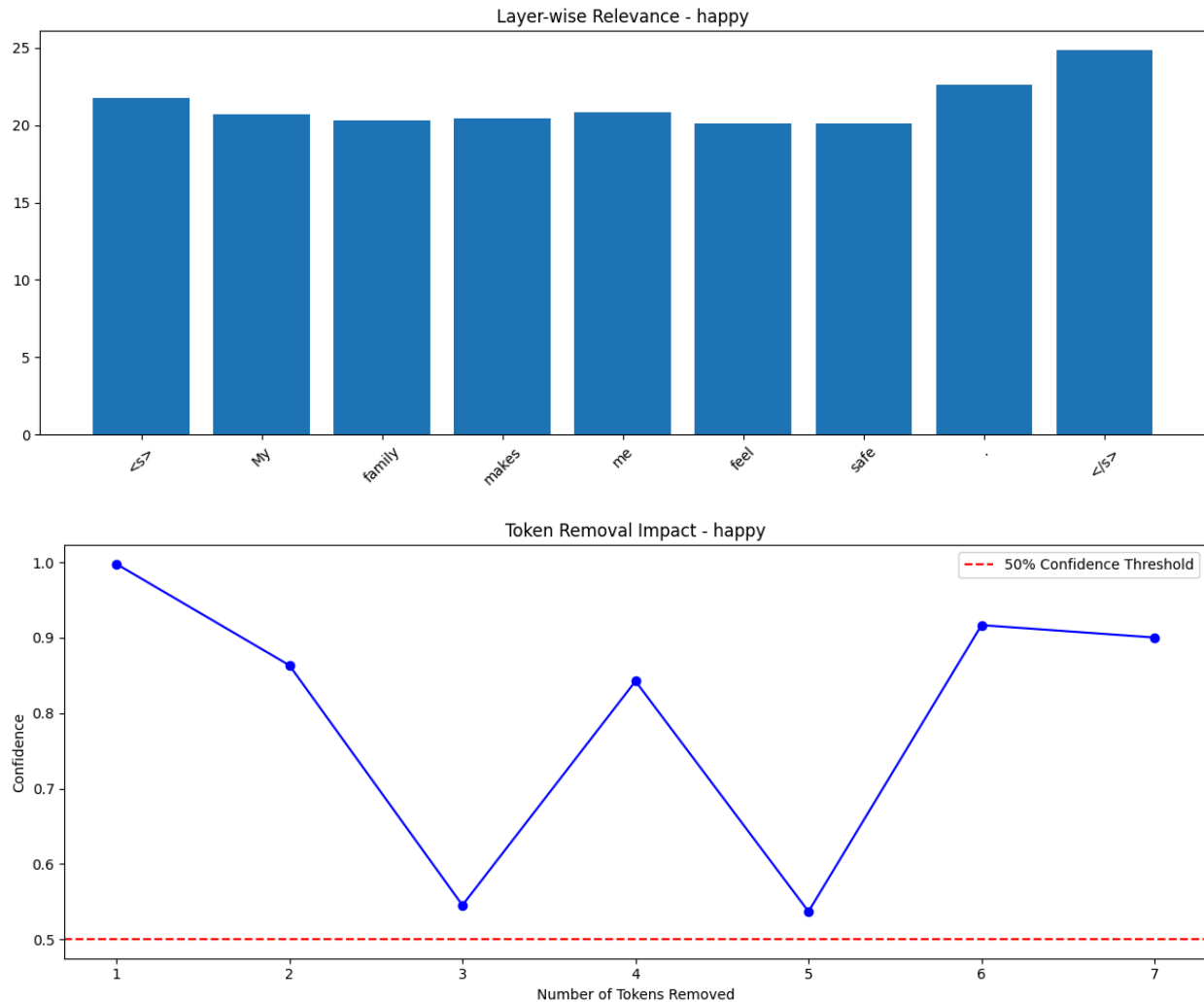
Sentence 5: "What a wonderful day full of happiness and laughter!"
Predicted emotion: happiness





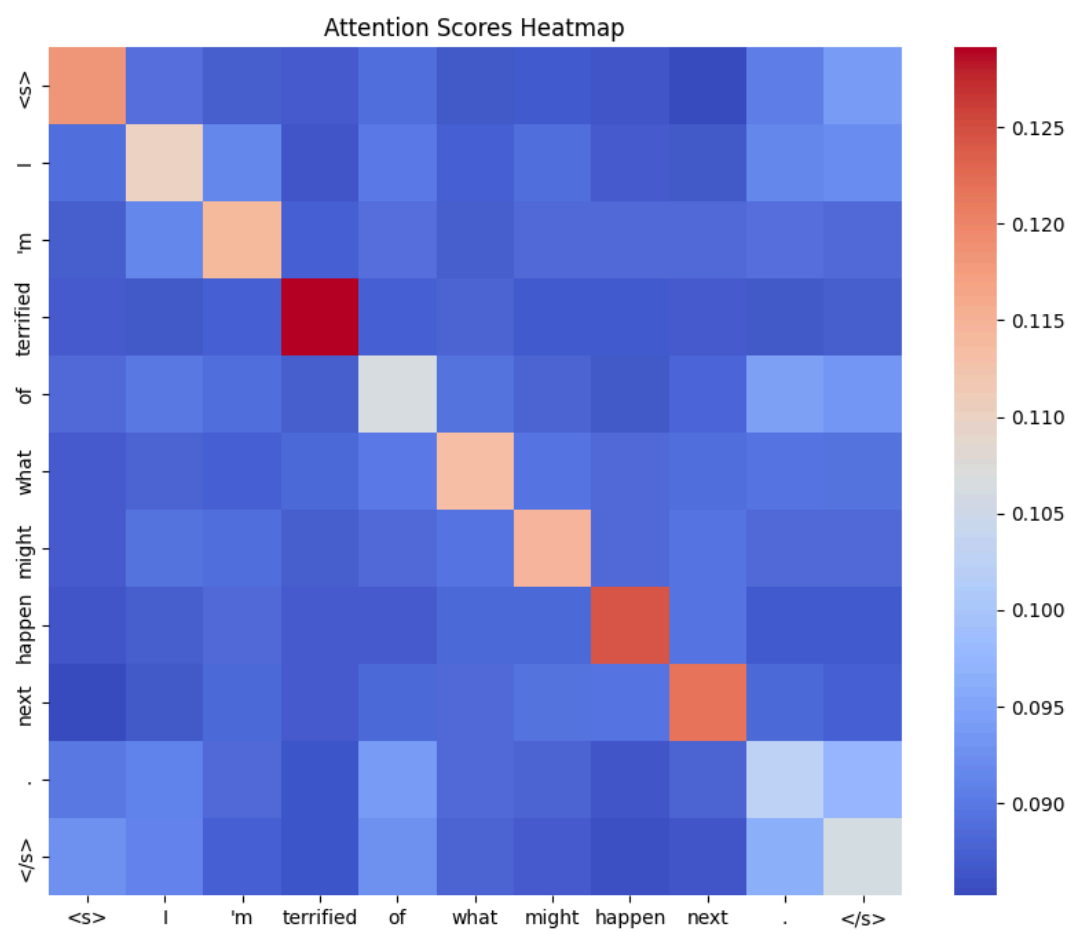
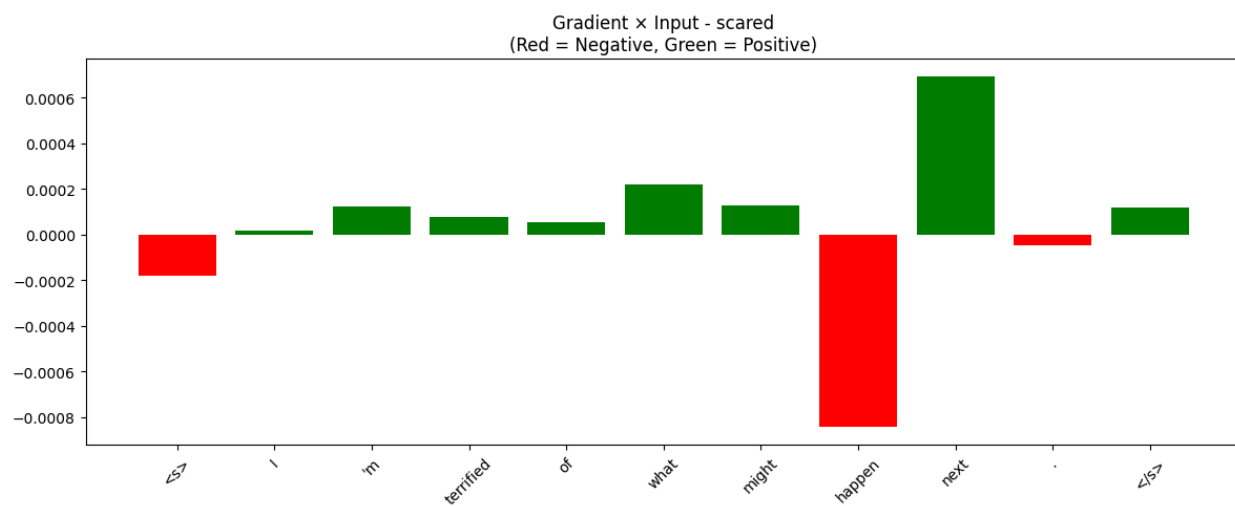
- Emotionally significant words like "wonderful" and "happiness" are expected to have high relevance, but the model assigns unexpected negative contributions to "happiness" and "laughter," favoring "wonderful" as the overall indicator of happiness.
- Irrelevant tokens like "and" and "!" influence the output negatively, while "happiness" and "laughter" receive lower-than-expected relevance scores.
- While "wonderful" strongly signals happiness, the negative contributions of "happiness" and "laughter" contradict intuition.
- Using LRP, the model better captures emotionally relevant tokens, with "happiness" and "laughter" receiving slightly higher relevance. However, "</s>" still has a disproportionate weight and also shows higher scores for emotionally neutral words such as "a" and "and".
- Compared to Gradient $\times$ Input, LRP places less emphasis on irrelevant tokens like "and" and "!", offering a more balanced attribution.
- A sharp confidence drop after removing a few tokens suggests strong reliance on specific words, while a gradual decline means the model considers multiple words. Here, the confidence remains high with some token removal, meaning the model is robust but

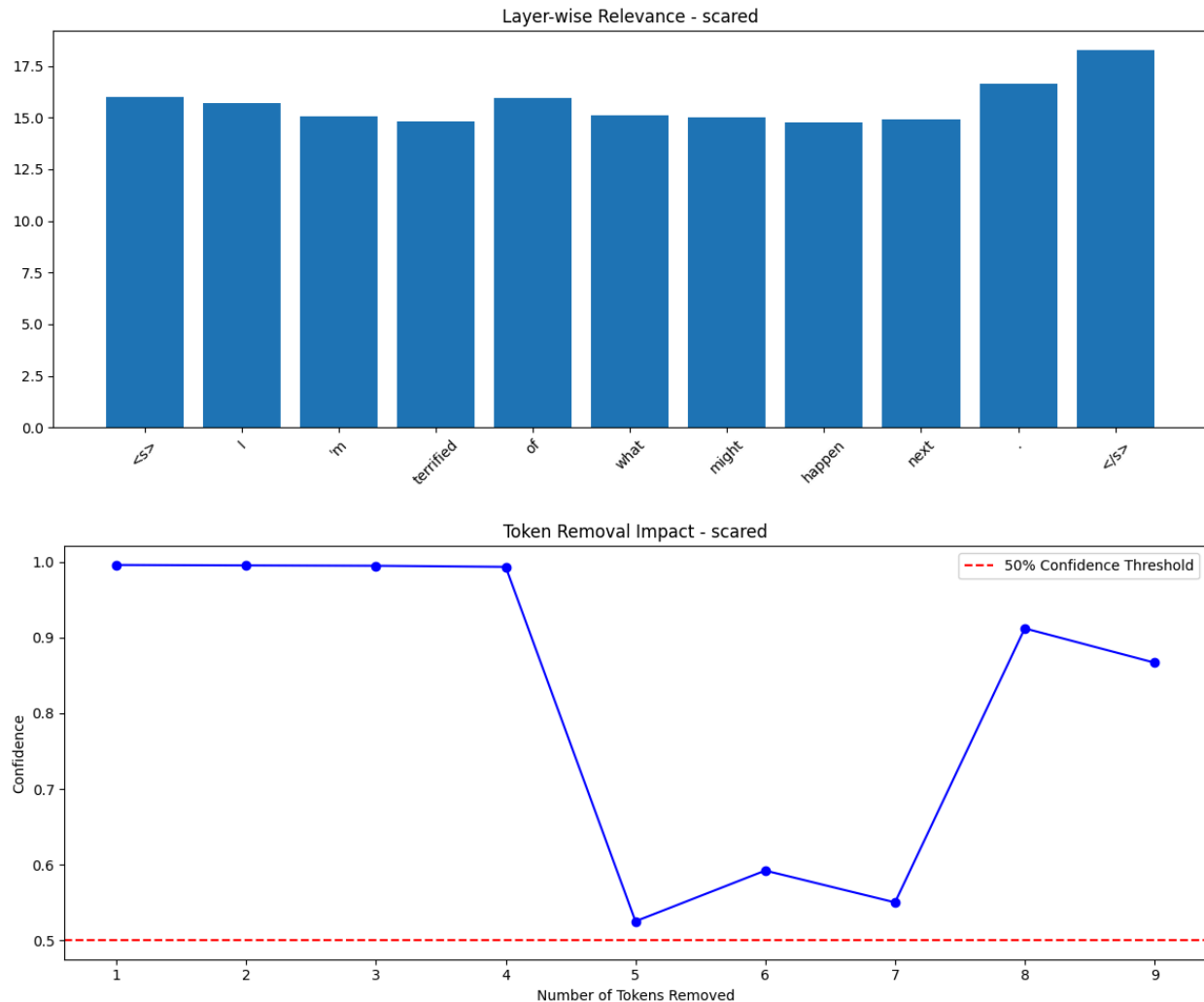




- The model correctly identifies "safe" and "feel" as strong indicators of "happiness." While less emotionally charged words also seem to be influencing the model's output, even though they almost have a neutral contribution, the model's overall interpretation aligns with my understanding. I would classify the sentence as expressing "happy" based on the presence of "safe" and "feel." The model heavily relies on the self-attention of the words. In the "Layer-wise Relevance" chart, "safe" and "feel" are not emphasized as much as in "Gradient $\times$ Input." In this case the gradient method is better because LRP does not diminish the importance of less relevant tokens. Removing the first three tokens ("My family makes") significantly drops confidence below 0.6. Removing "me" further drops confidence to near chance level, highlighting its role. However, removing "feel" and "safe" then sharply increases confidence, suggesting the remaining context ("My family makes") somehow signals "happy" to the model, despite being incomplete.

Predicted emotion: scared



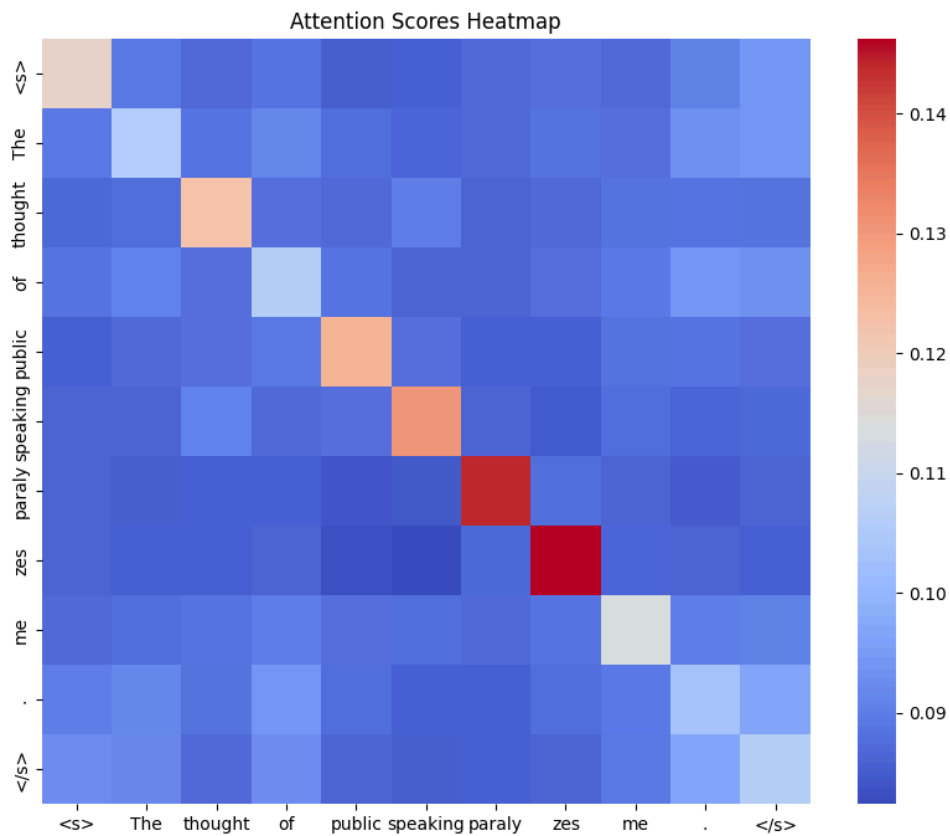
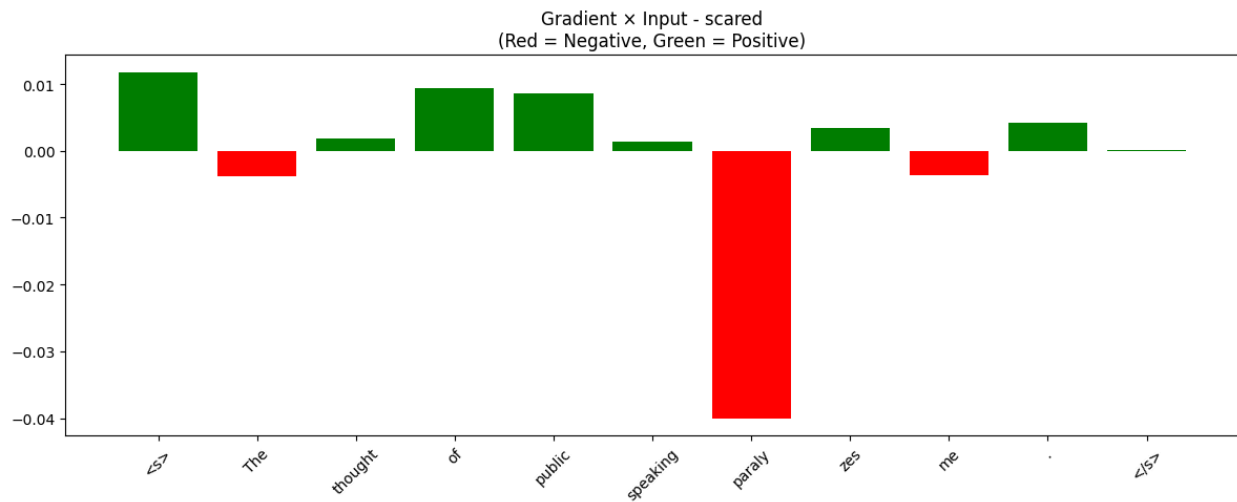


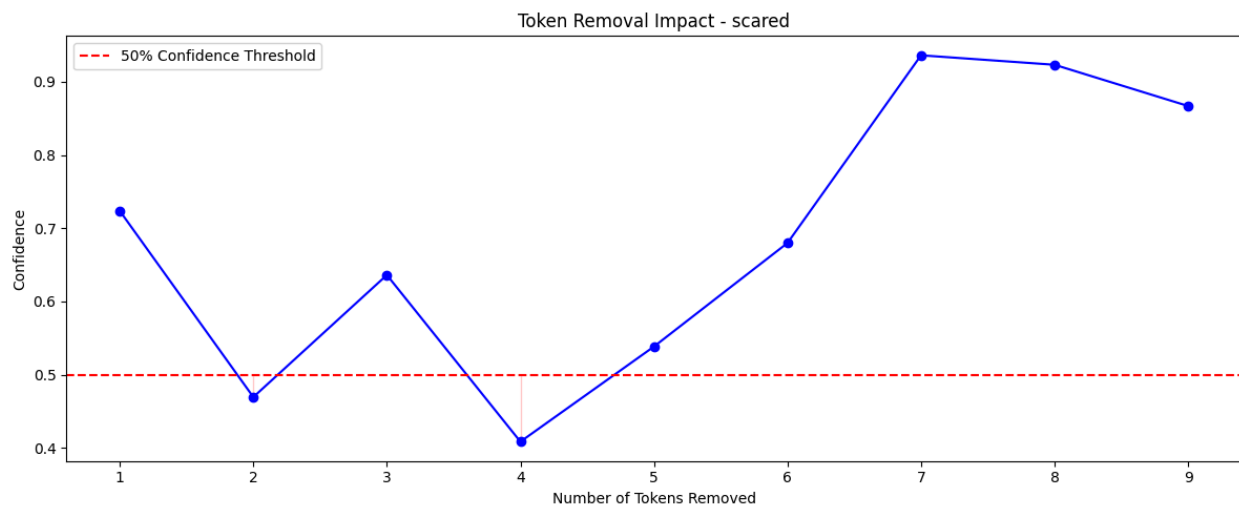
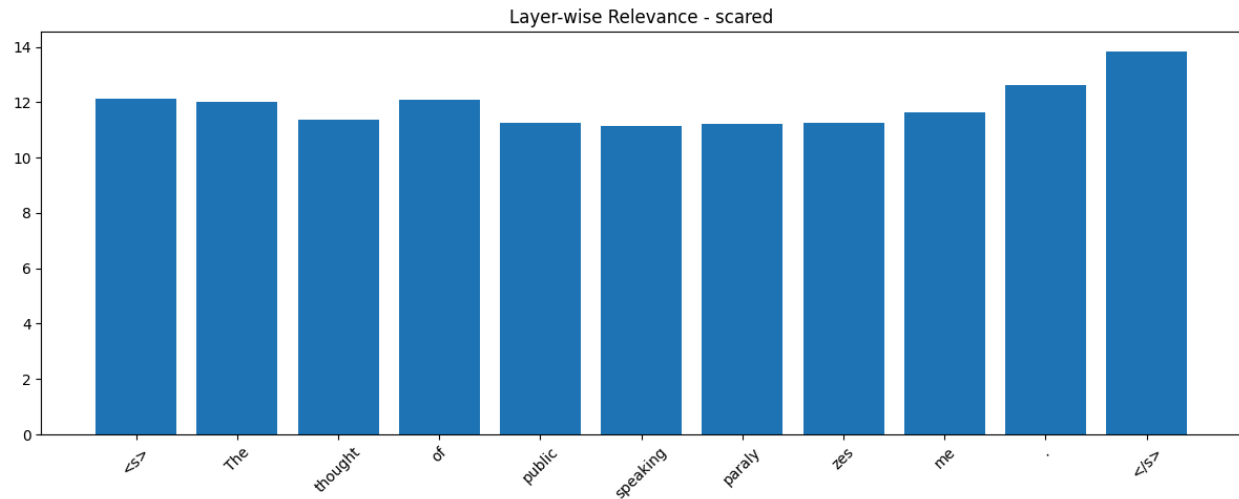
- I agree that words like "terrified" and "happen" align well with the model's scoring in Gradient $\times$ Input. The model assigns the highest positive relevance to "next," so it strongly associates fear with the sense of something imminent, while "terrified" also reinforces the emotion as expected. Surprisingly, "happen" has the most negative relevance, which is counterintuitive since it relates to potential frightening events.
- The model relies heavily on sentence structure when predicting "scared," as shown by the high relevance of start and end tokens. Unlike Gradient $\times$ Input, LRP spreads relevance more evenly, suggesting that while "happen" isn't a direct fear indicator, it still helps shape the overall context. LRP also shows how each word contributes to the prediction based on the full sentence, with "terrified" remaining a key but not dominant indicator of fear. The attention heatmap shows the model heavily prioritizing the word "terrified" as a direct indicator of "scared,". The drop in confidence after removing key tokens suggests that the model depends heavily on specific emotionally charged words "terrified" and "happen." If confidence decreases gradually, it means the model distributes its understanding across multiple words rather than just a few critical ones.

The counterintuitive confidence increase when removing a specific token suggests that some words may introduce noise rather than aid classification.

Sentence 8: "The thought of public speaking paralyzes me!"

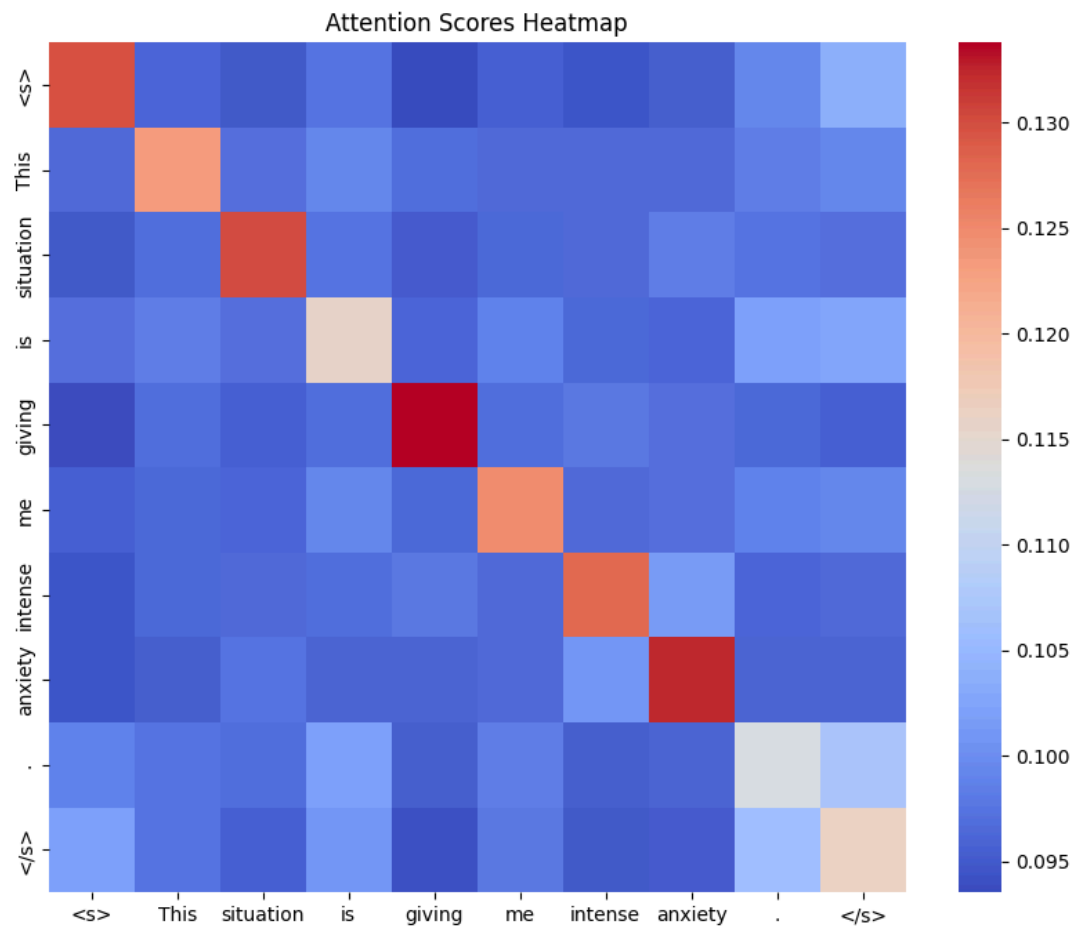
Predicted emotion: scared

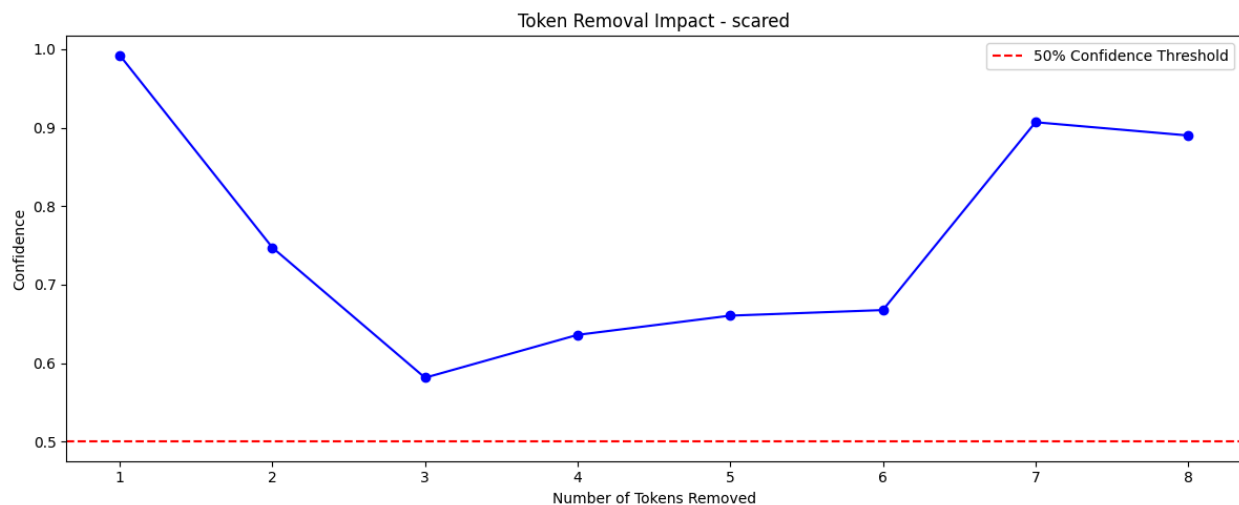
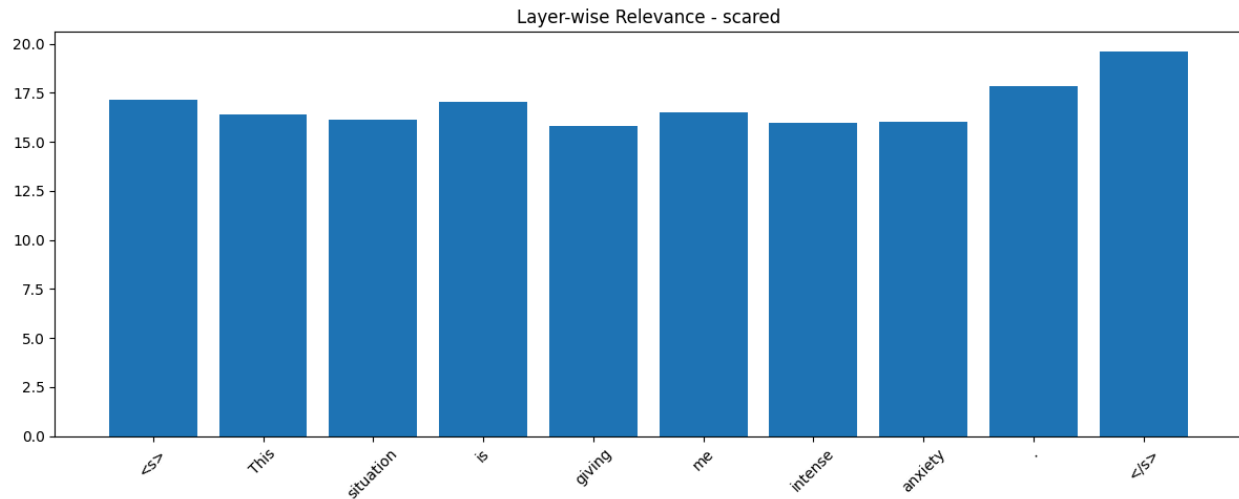




- "Paralyzes" is a very strong word associated with fear, and it has a high magnitude of influence. However, it's a negative influence, meaning it pushes the model away from classifying the sentence as "scared." This is counterintuitive. Other words like "thought," "public," and "me" have small positive contributions, meaning they slightly increase the probability of "scared", which I can agree with. I am not sure if I would assign this emotion since the best word to convey fear is actually not helping the model at all. This heatmap visualizes the self attention of words like "paralyzes". While "paralyzes" is still considered, its relevance isn't as highlighted by LRP as in the "Gradient $\times$ Input" explanation. The model starts with high confidence above 0.7, but removing one or two tokens causes a significant drop below 50%. As more tokens are removed, confidence fluctuates, showing that some words have a greater impact than others. Removing 7 or 8 tokens results in peak confidence, suggesting the remaining words are key for predicting "scared."

Predicted emotion: scared

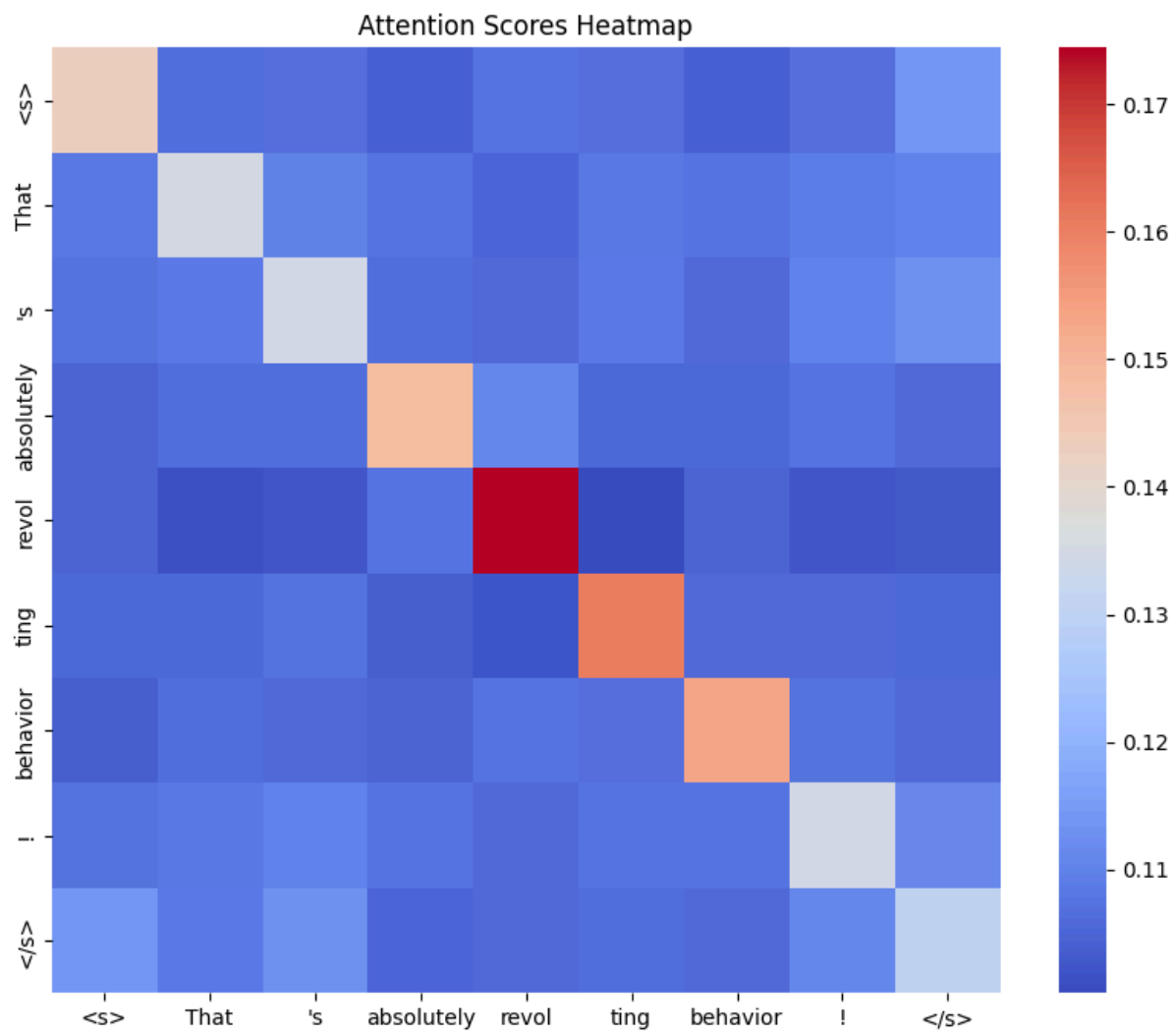
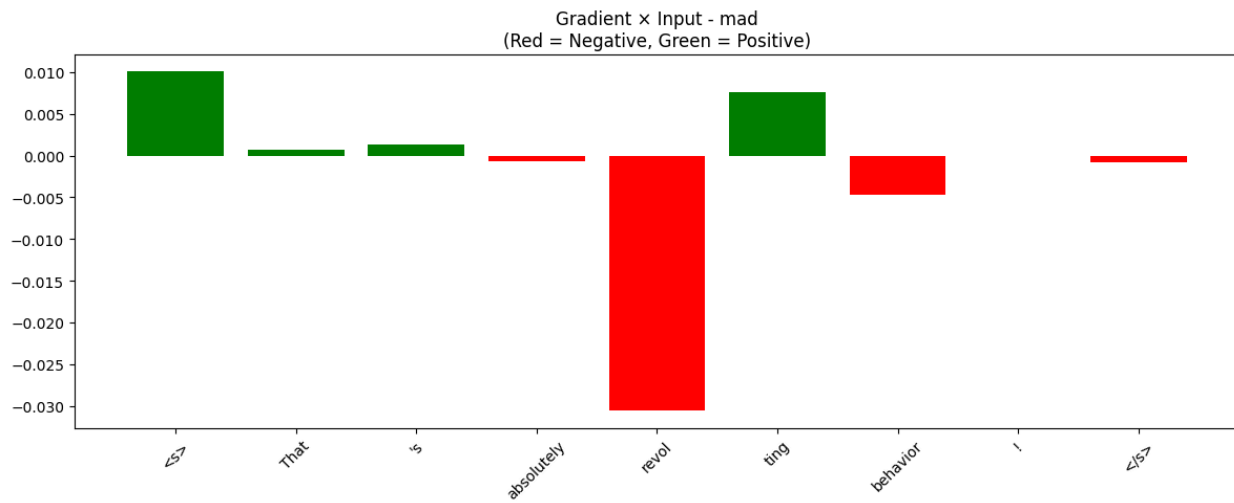


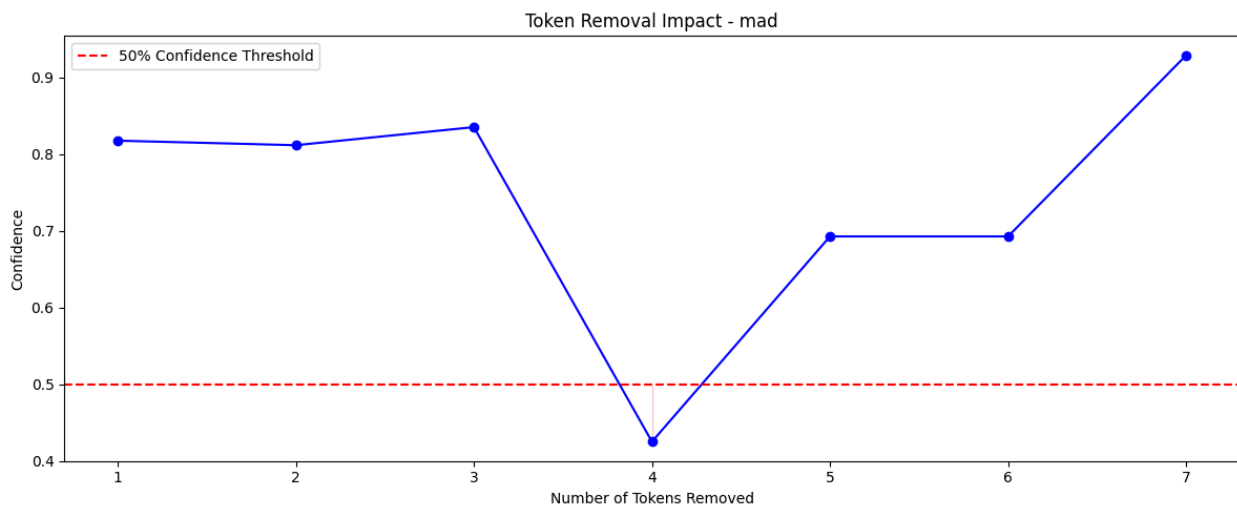
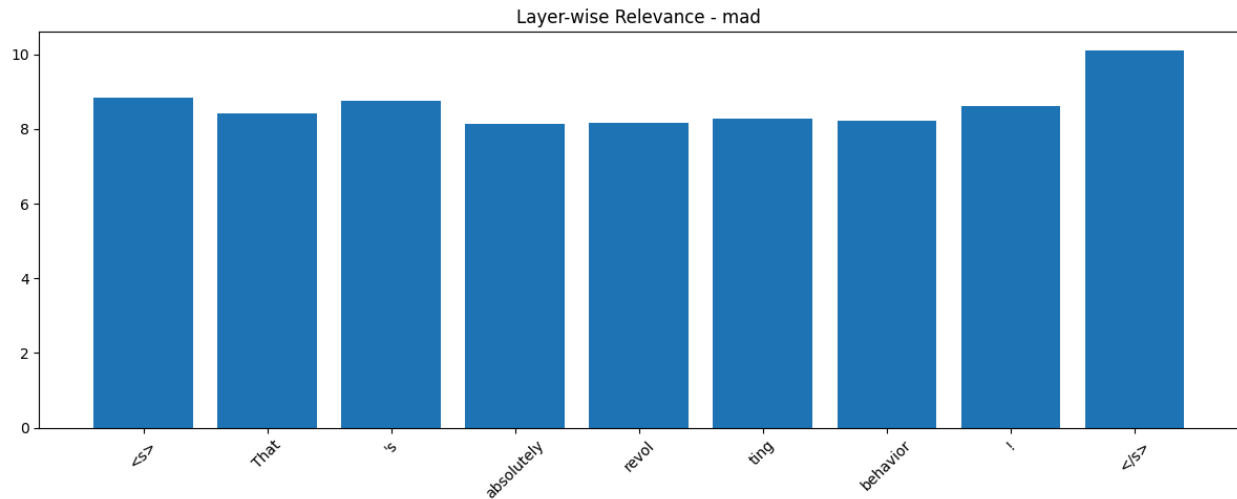


- The emphasis on "intense" and "anxiety" aligns with my expectations. These words are key to understanding the emotional tone. The negative influence of "me" and "this" is a bit counterintuitive. I would still assign this emotion. In the "Layer-wise Relevance" chart, "intense" and "anxiety" are not emphasized as much as in "Gradient $\times$ Input." All tokens are considered relatively important. The least valuable token removal analysis shows that the model is very sensitive to removing even the first token, but it is still robust above threshold. There is a sudden rise after removing the 7th token which suggests the model is highly reliant on word combinations.

Sentence 10: "That's absolutely revolting behaviour!"

Predicted emotion: mad

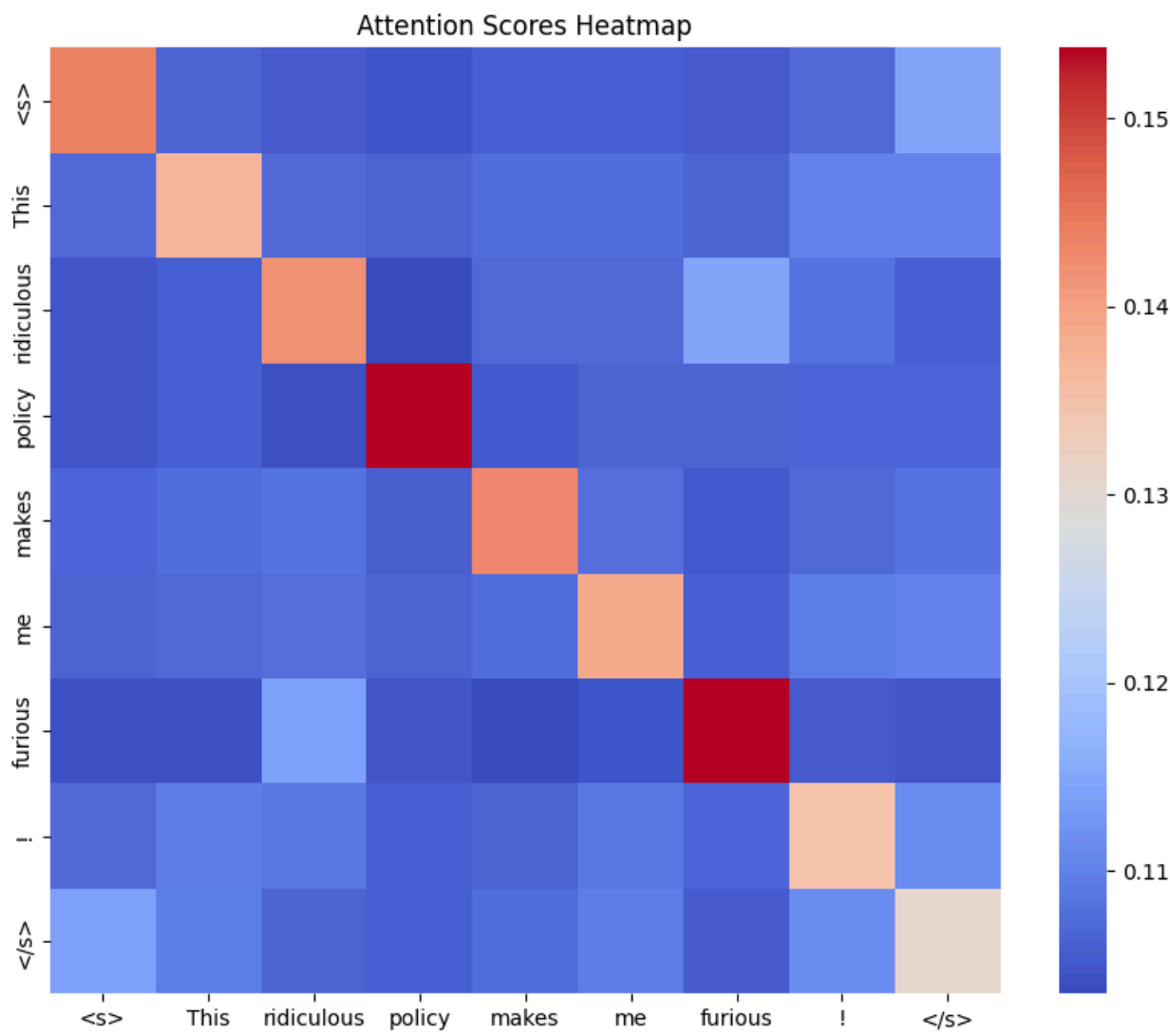
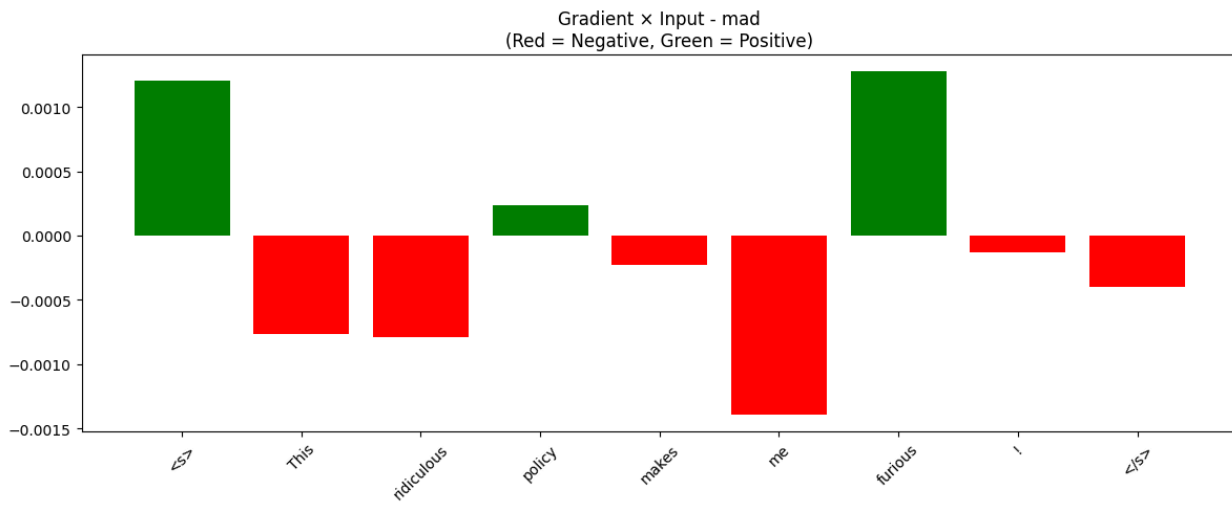


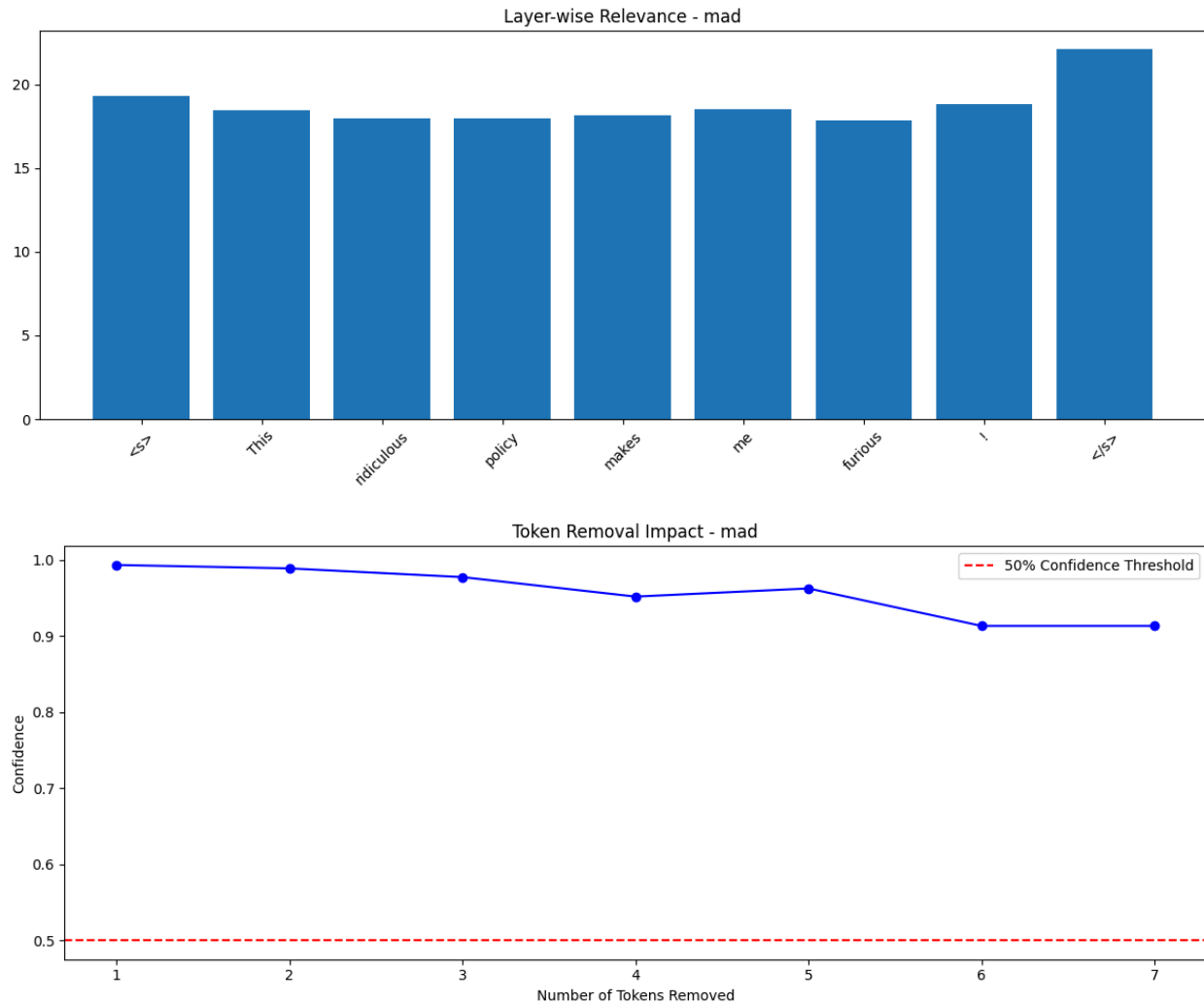


- I would expect "revolting" to strongly increase the probability of "mad." The model assigns "revolting" a very high negative relevance. This means it strongly reduces the probability of the sentence being classified as "mad." which is counterintuitive. I would not assign this emotion as only the "that's" and "ting" seem to have a positive contribution towards a proper classification. Great self-attention on the word "revolting" In the "Layer-wise Relevance" chart, "revolting" is not emphasized as much as in "Gradient $\times$ Input.", all words seem to have similar contributions. It again emphasizes greatly the end of the sentence, which may suggest the need of the model for context. The graph shows the model initially confident in "mad," but removing four tokens ("revolting") drops confidence sharply, revealing its importance; confidence fluctuates with further removals, then goes back up when only "behavior", "!" remain. the model finds "mad" even in minimal context, despite initial sensitivity to the full sentence.

Sentence 11: "This ridiculous policy makes me furious!"

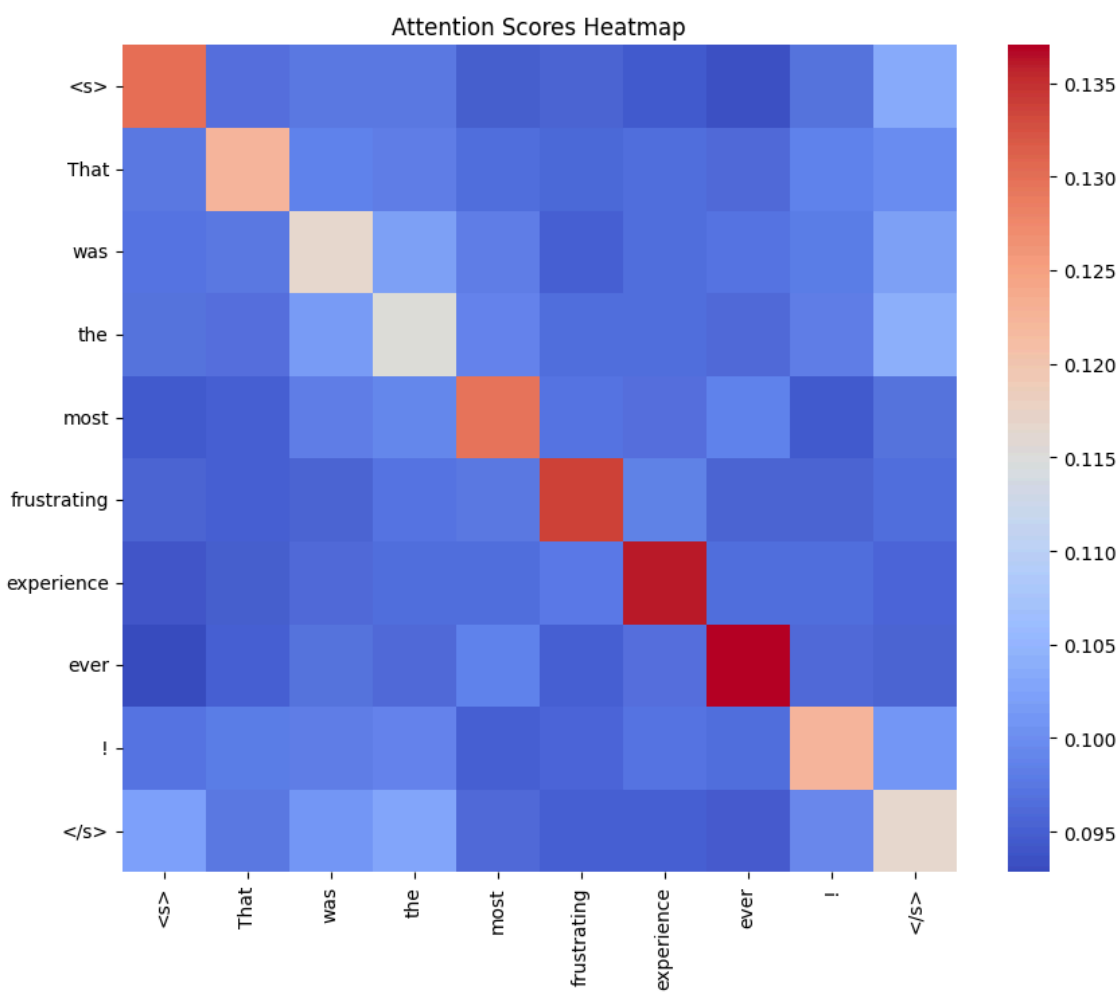
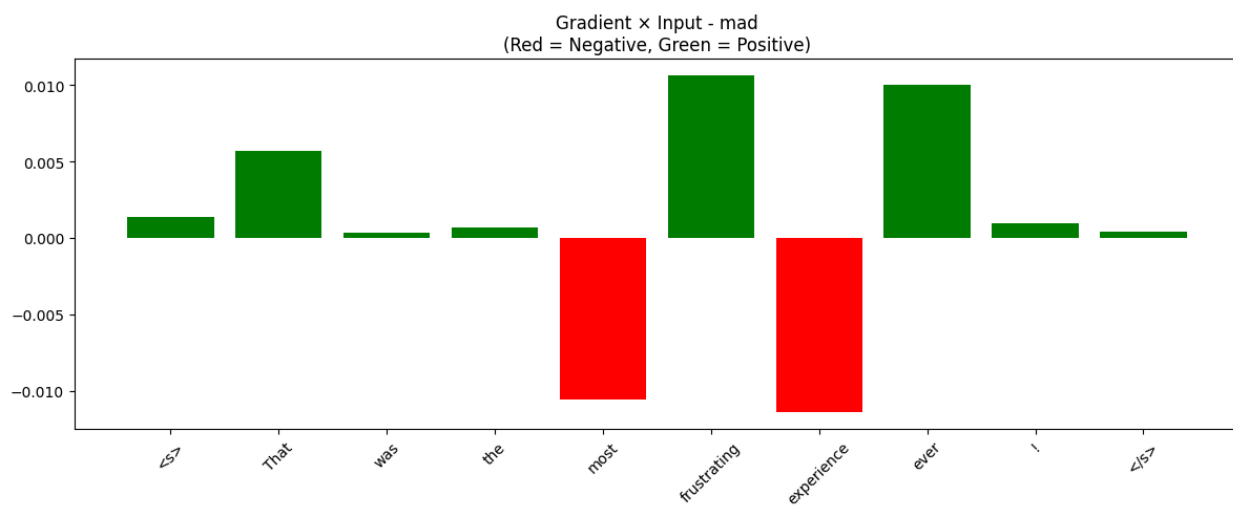
Predicted emotion: mad

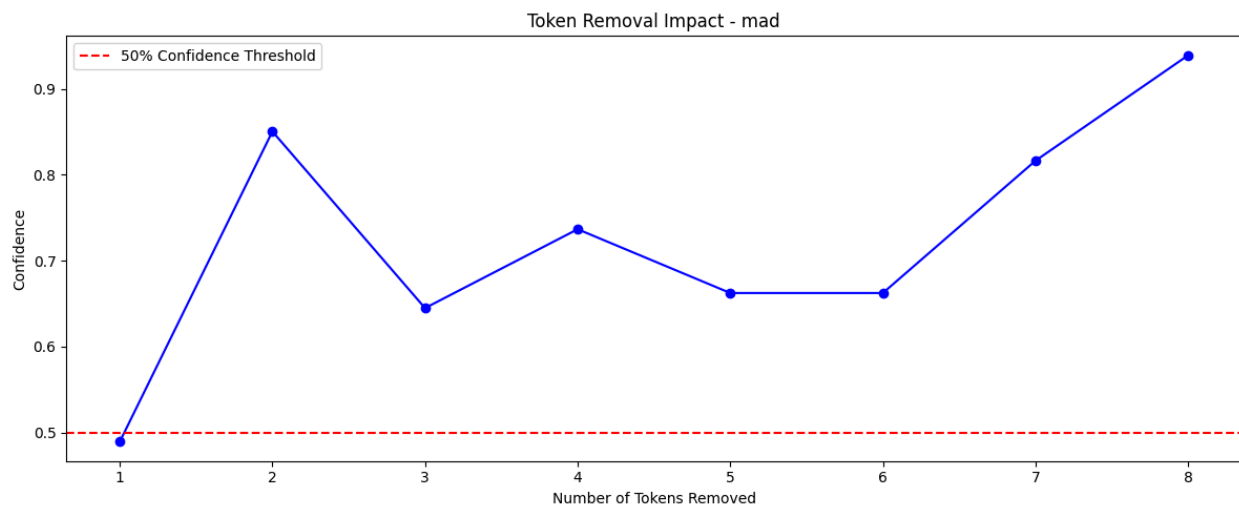
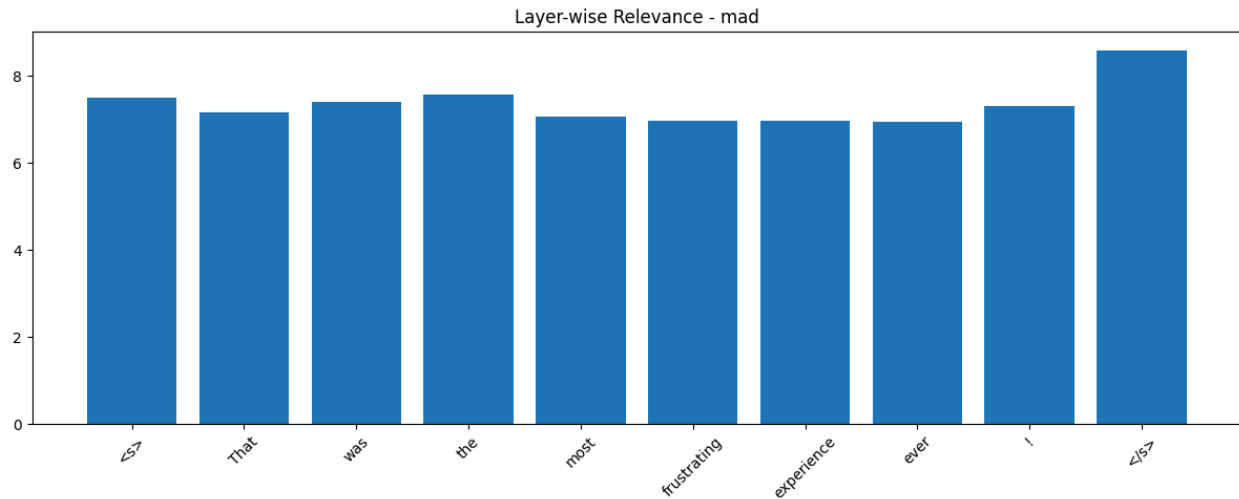




- The model correctly identifies "furious" as strongly indicating "mad," but surprisingly, it interprets "ridiculous" as reducing the probability of "mad," which is counterintuitive. Similarly, "me" is also assigned a strong negative influence, which doesn't align with my understanding. While less emotionally charged words like "this," "policy," "makes," and "I" also have some influence, they don't seem as relevant. I would still classify the sentence as expressing "mad" due to the clear presence of "ridiculous" and "furious." This heatmap shows each word attends strongly to itself, as evidenced by the red/orange diagonal. "Layer-wise Relevance" chart assigns similar relevance scores to all words. Emotionally relevant tokens like "ridiculous" and "furious" are not emphasized more than other tokens, and irrelevant tokens are not less emphasized, so the gradient aligns more closely with my human understanding. The confidence only slightly decreases with each token removal, remaining consistently above 0.9. This suggests the model finds strong indicators of "mad" throughout the sentence and isn't reliant on word combinations.

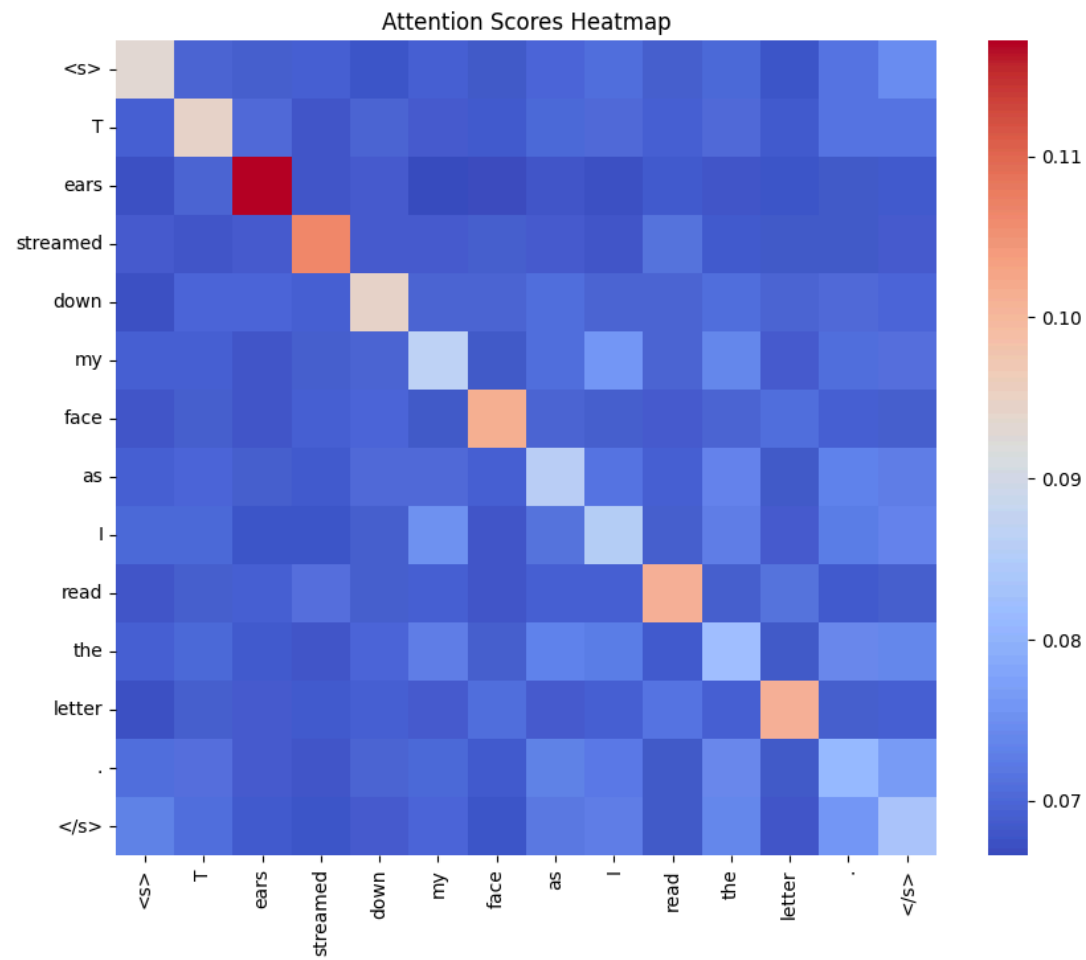
Predicted emotion: mad

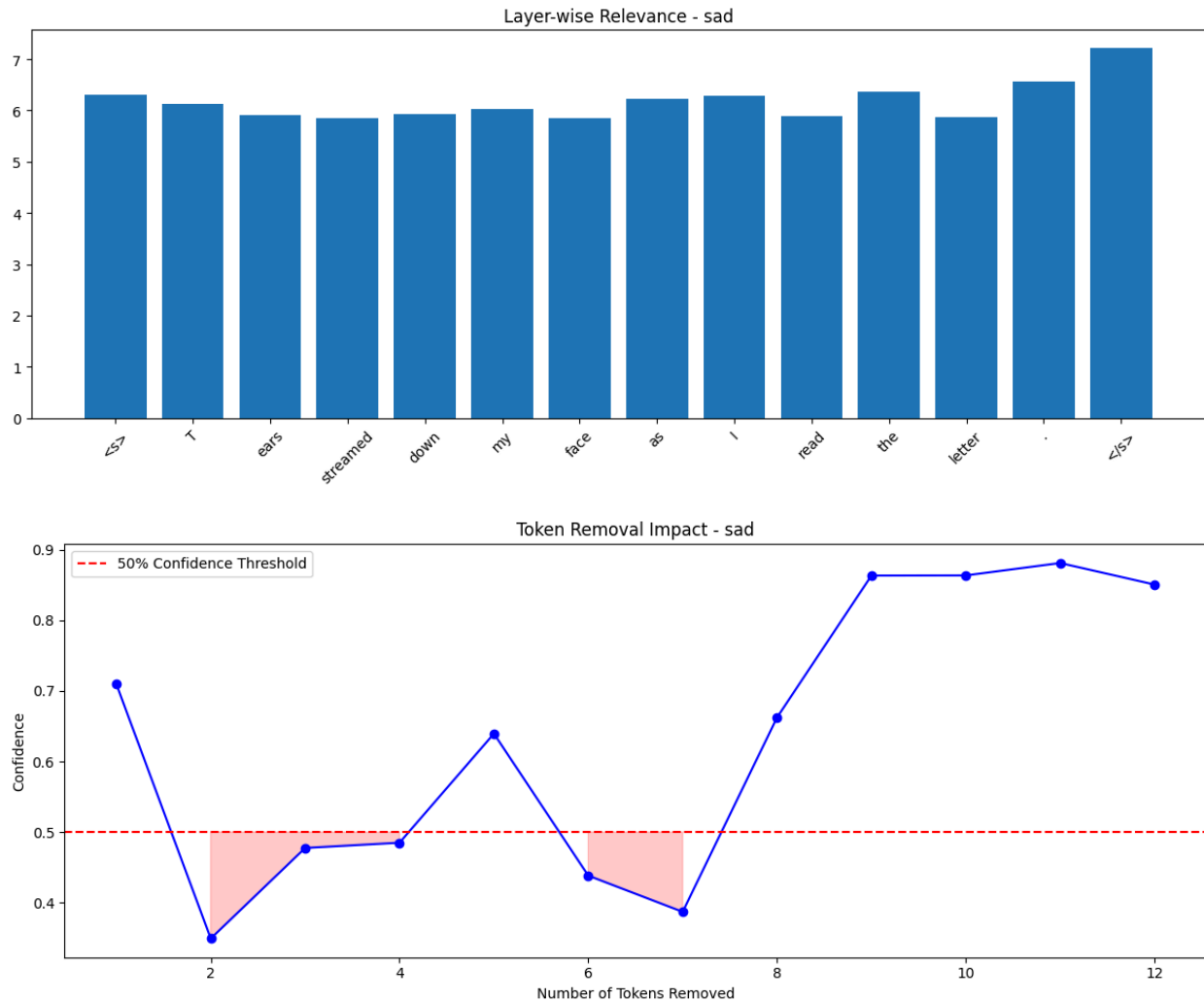




- The model assigns "frustrating" a very high positive relevance. This means it strongly increases the probability of the sentence being classified as "mad." which is also how I would identify this sentence for this emotion. "Experience" and "most" have strong negative influences, which are counterintuitive. "That," "was," "the," "ever," and "I" have small positive influences, but aren't strongly indicative of "mad" to me. The model is relying on self-attention, where each token is primarily focused on itself. Strong attention on a token suggests that the model recognizes its emotional importance. The "Gradient $\times$ Input" method highlighted "frustrating" as having a strong positive influence and "experience" and "most" as having strong negative influences. In the "Layer-wise Relevance" chart, these tokens have similar relevance to the emotionally charged ones. The first token drastically reduces the model's confidence in classifying the sentence as "mad. More removals show fluctuating confidence, indicating sensitivity to specific word combinations, but removing all but the last two shows that the model finds a strong "mad" emotion in those remaining words.

Predicted emotion: sad

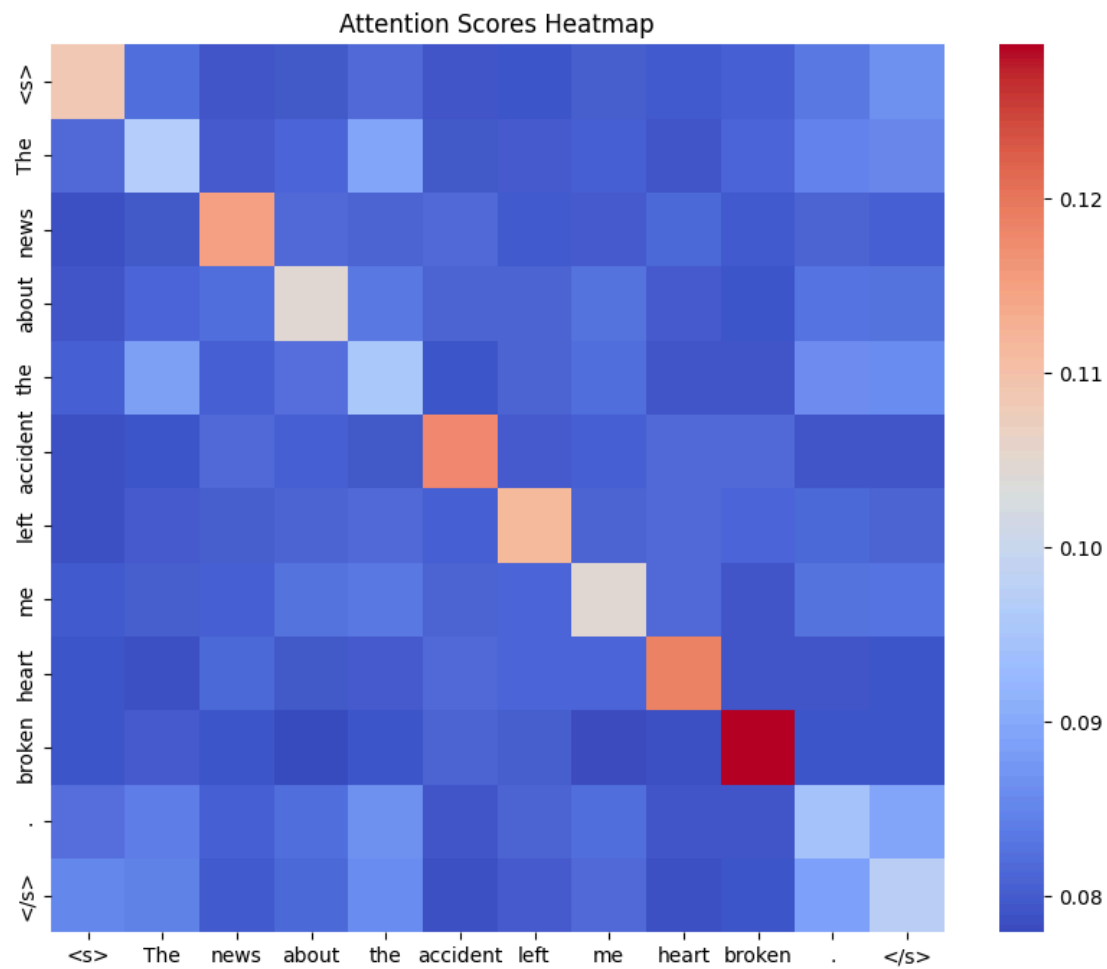
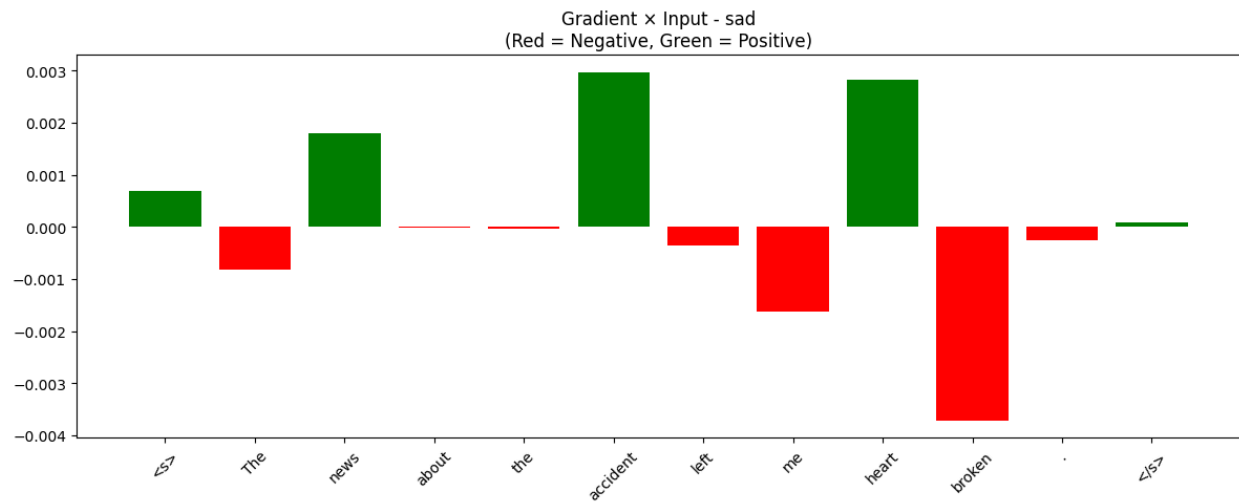


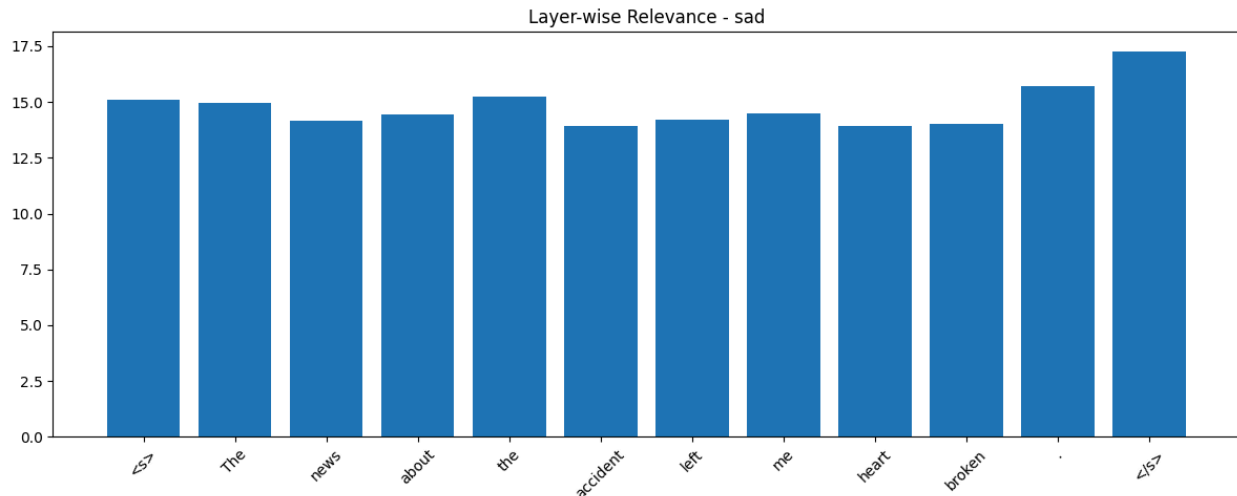


- The token “ears” has a really strong negative contribution which I can agree with. It confuses the model to identify the correct emotion, this may be a tokenization problem as the word “tears” would have been a lot more suggestive with this emotion of sadness. The words “streamed” and “down” have expected positive contributions given the context. The words “my”, “face” and “letter” have negative contributions which are counter intuitive. The model relies on self-attention of the words. In the "Layer-wise Relevance" chart, "tears" and "streamed" are not emphasized as much as in "Gradient × Input." All tokens are considered relatively important, which may be more appropriate for this context. Removing the first few tokens significantly drops the model's confidence in classifying the sentence as "sad,"; subsequent removals cause fluctuating confidence, suggesting sensitivity to specific word combinations. Removing most of the tokens leads to a sharp increase in confidence, suggesting the model finds strong "sad" signals in the remaining words suggesting the first tokens were perceived as noise.

Sentence 14: "The news about the accident left me heartbroken!"

Predicted emotion: sad



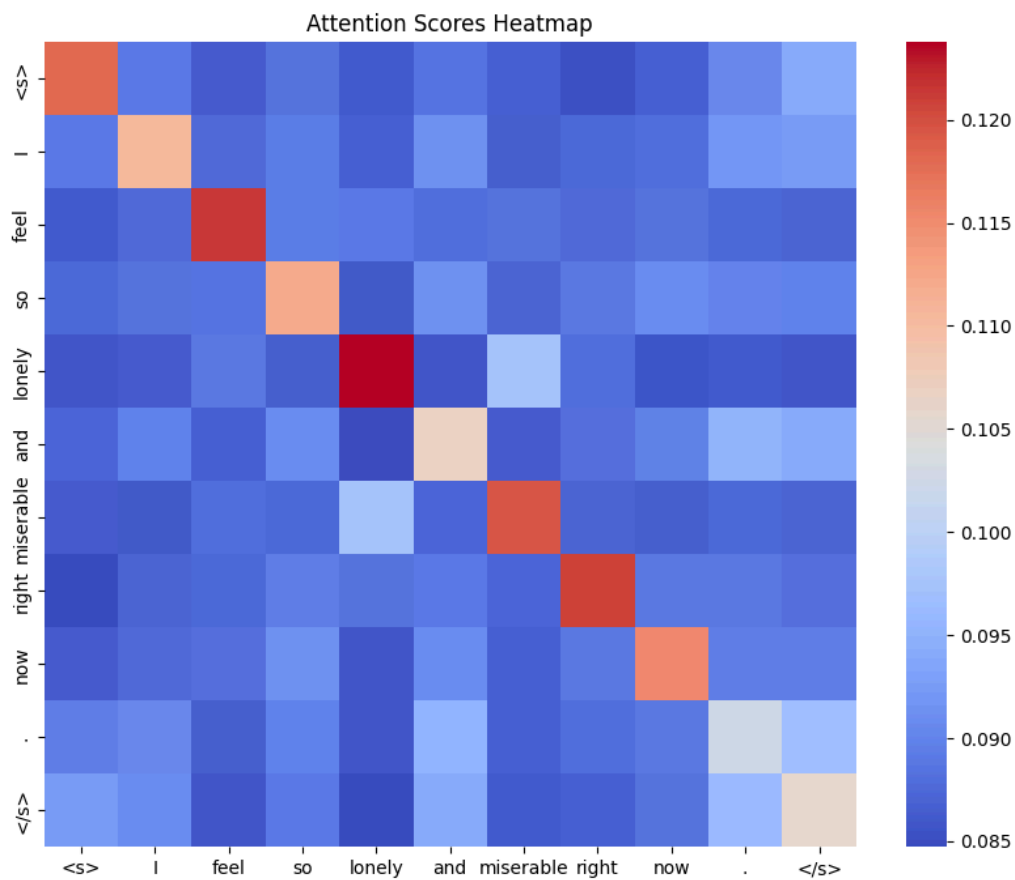
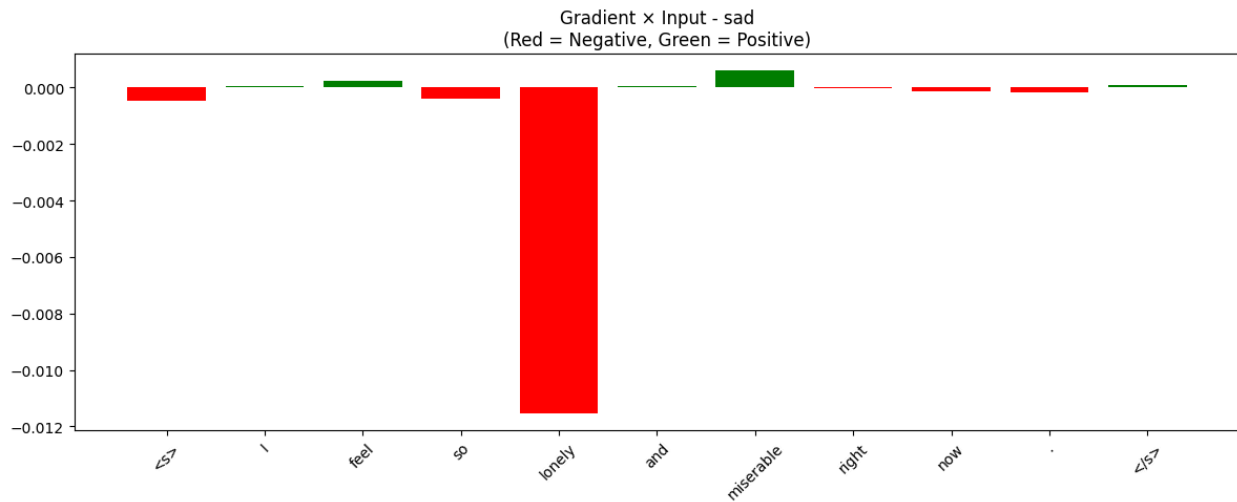


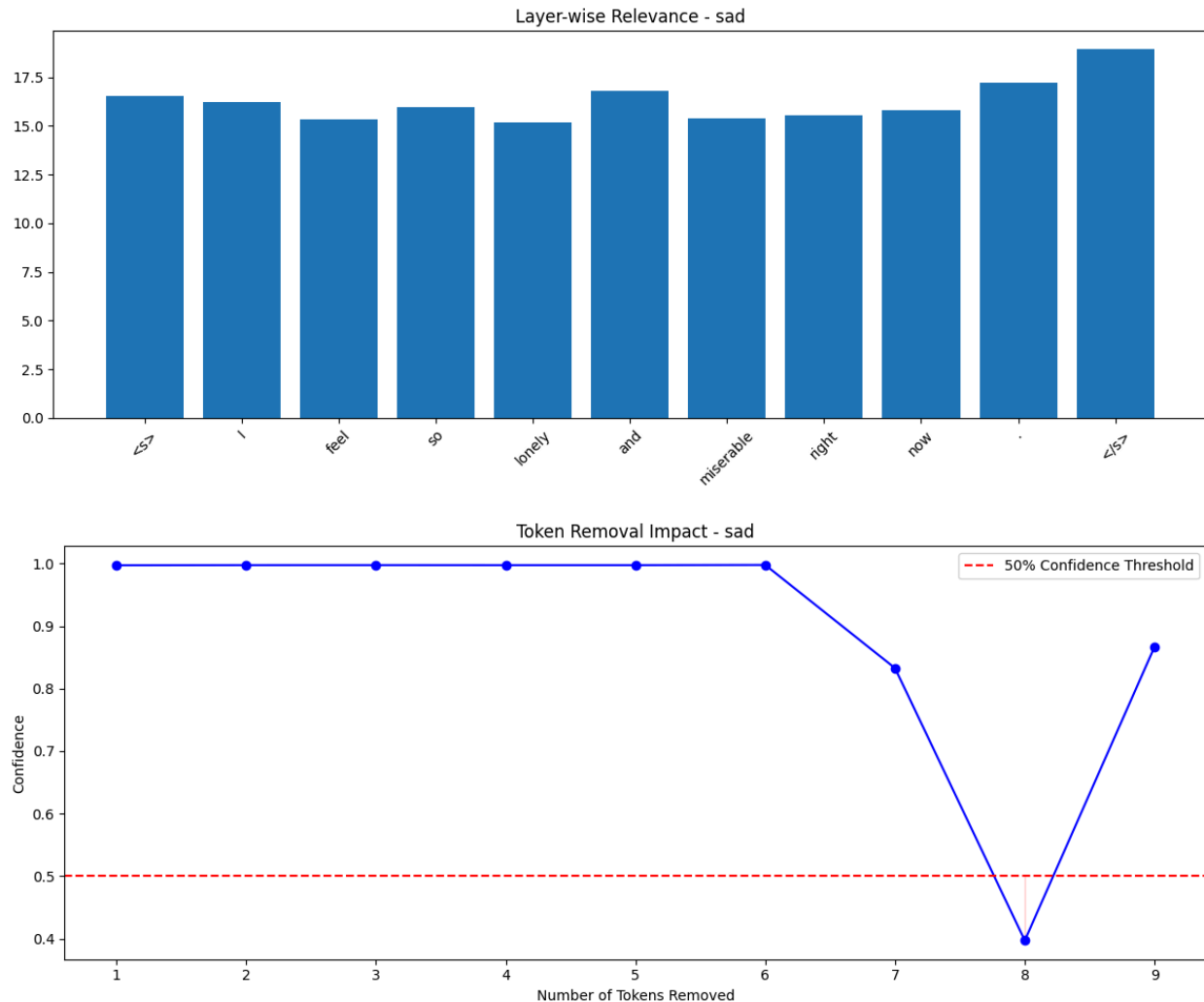
- The model correctly identifies "accident" as indicating "sad," but its interpretation of "heart" has an expected positive contribution as it emphasizes a more dramatic context but "broken" is completely counterintuitive and doesn't align with my understanding. Less emotionally charged words also seem to be influencing the model's output, as all tokens have a minimal input but an input nonetheless. But based on the highlighted tokens alone I would still identify this emotion. The model still relies on self-attention of the words. In LRP all tokens have moderate relevance scores (around 14-15), suggesting the model considers all parts of the sentence to contribute to the "sad" classification. The start and end tokens (<s> and </s>) have slightly higher relevance, which is common in language models. The "Gradient $\times$ Input" method showed some impact from less emotionally charged tokens like "the," "about," "left," and ".". In the "Layer-wise Relevance" chart, these tokens have similar relevance to the emotionally charged ones, in this context the gradient seems better. Model's initial confidence drops significantly after removing three tokens ("about"), highlighting its importance for this emotion; further removals lead to fluctuating confidence, with another drop at six removals ("left"); yet, removing most tokens results in a sharp confidence increase, so the model still finds this

emotion with few words. This also shows the dependent word relationships that can become noise if that relation is broken.

Sentence 15: "I feel so lonely and miserable right now."

Predicted emotion: sadness

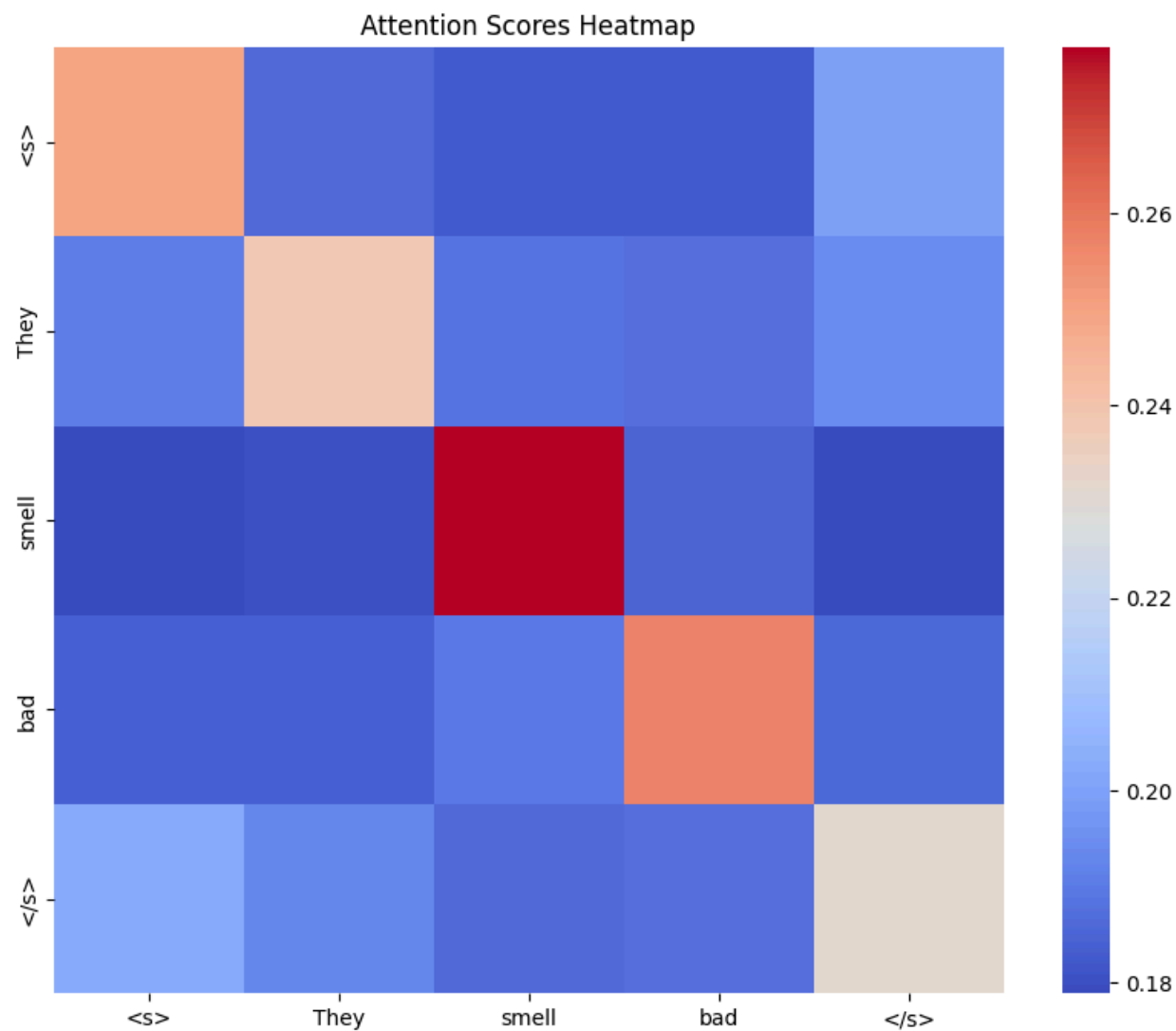


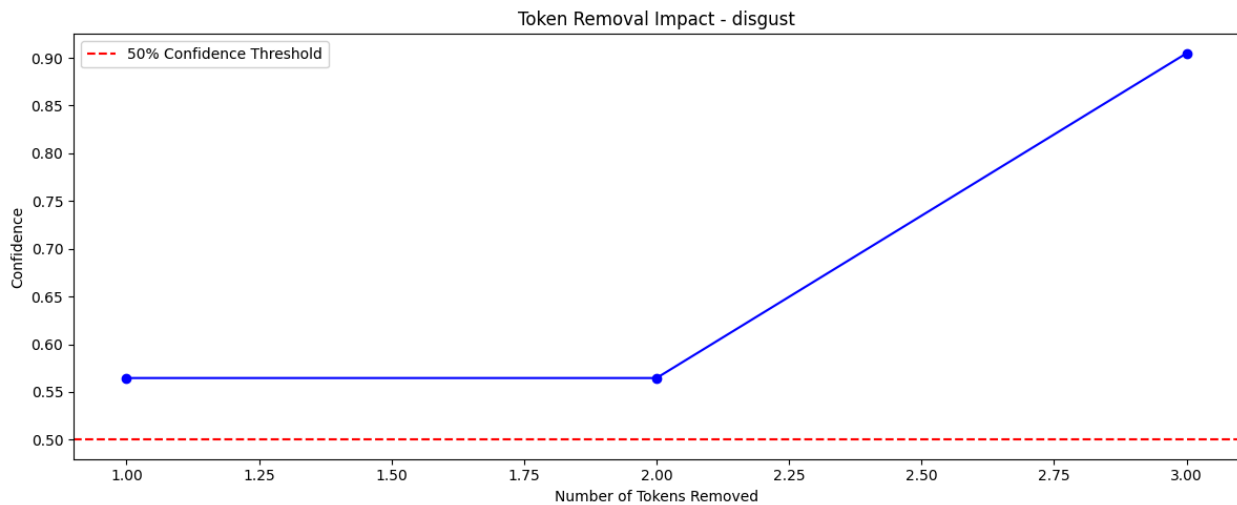
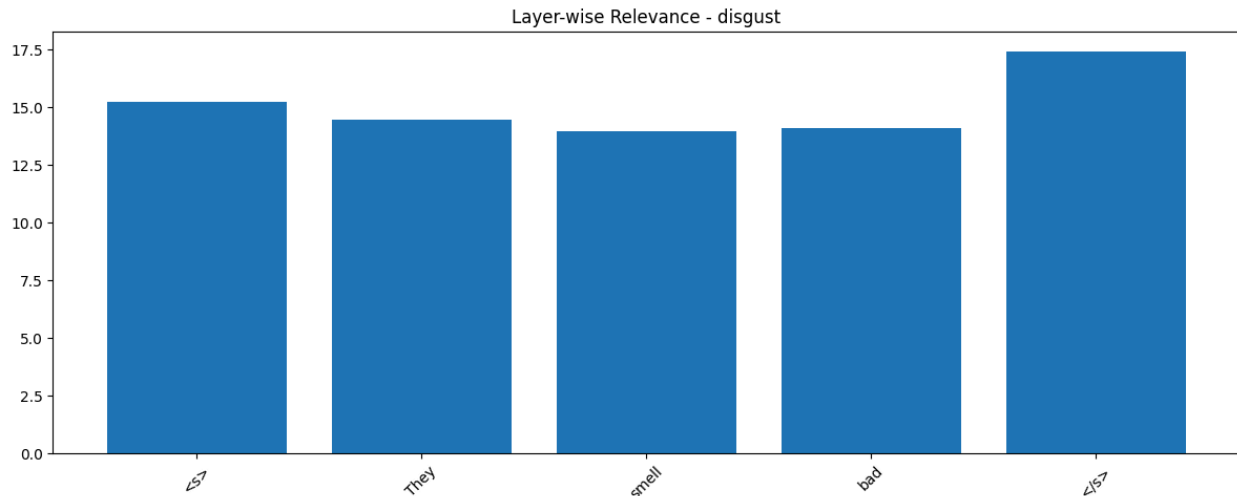


- The model correctly identifies "miserable" as indicating "sad," but its interpretation of "lonely" is completely counterintuitive and doesn't align with my understanding. Less emotionally charged words also seem to be influencing the model's output, but in a confusing and fluctuating manner. Despite the model's counterintuitive interpretation of "lonely," I would still classify the sentence as expressing "sad" due to the presence of "lonely" and "miserable." The model still relies on self-attention of the words but the whiter spots show that the model also captures some relationships but weakly. In the "Layer-wise Relevance" chart, "miserable" and "lonely" are not emphasized as much as in "Gradient $\times$ Input." Irrelevant tokens are not less emphasized in the "Layer-wise Relevance" chart. They have a similar level of relevance as the emotionally charged words. The "Layer-wise Relevance" chart, while not highlighting the key emotional words, might provide a more balanced view. The model has a robust high confidence until the 8th token ("now") and then spikes back up, This suggests the model was relying on the co-dependent relationship between words to make its prediction.

Sentence 16: "They smell bad"

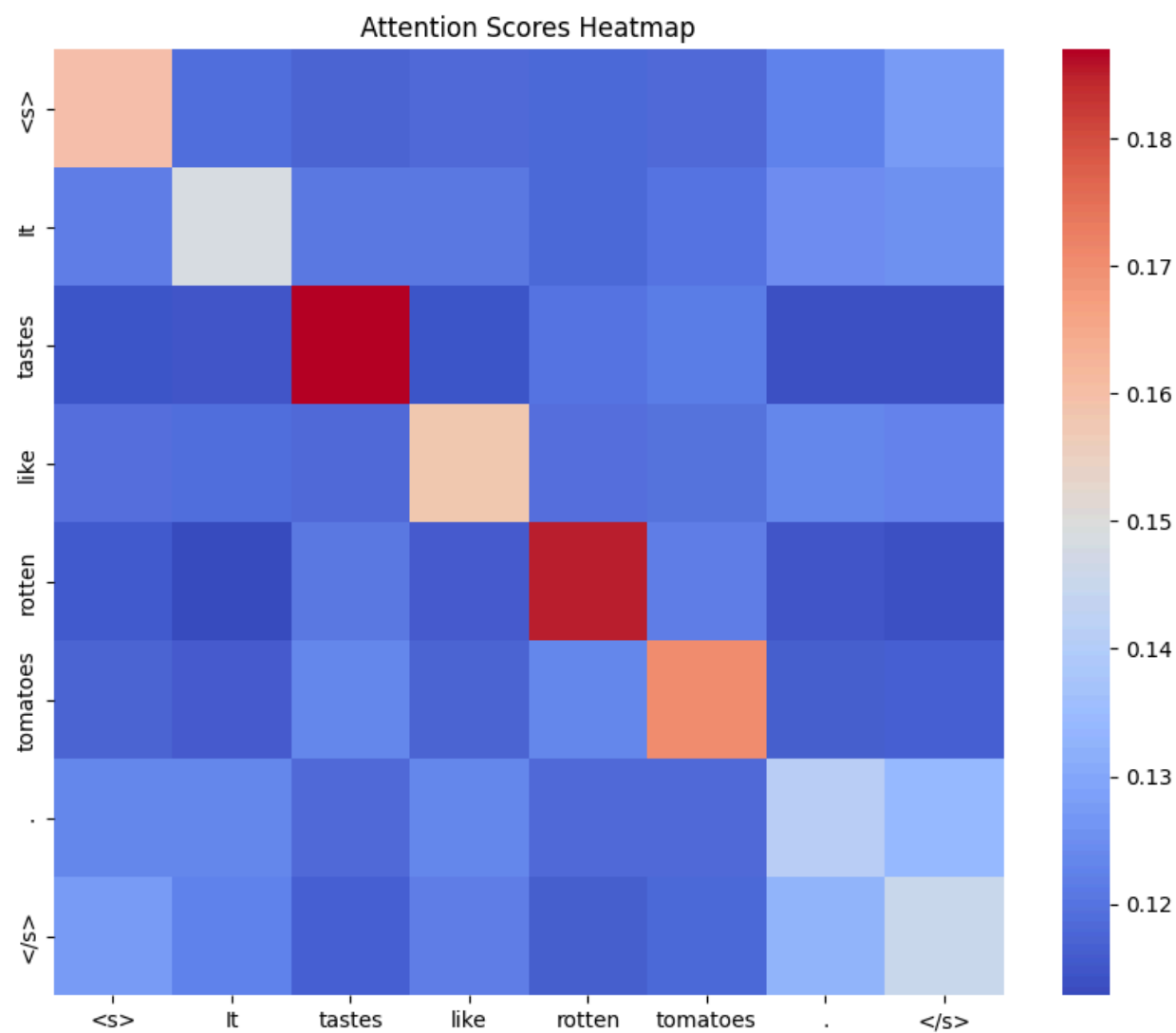
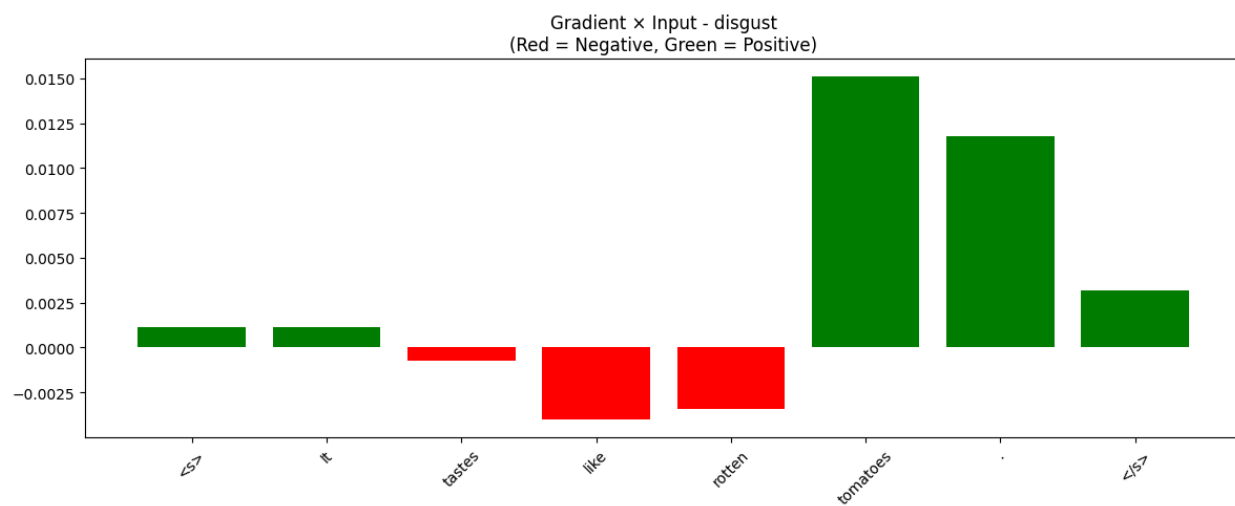
Predicted emotion: disgust

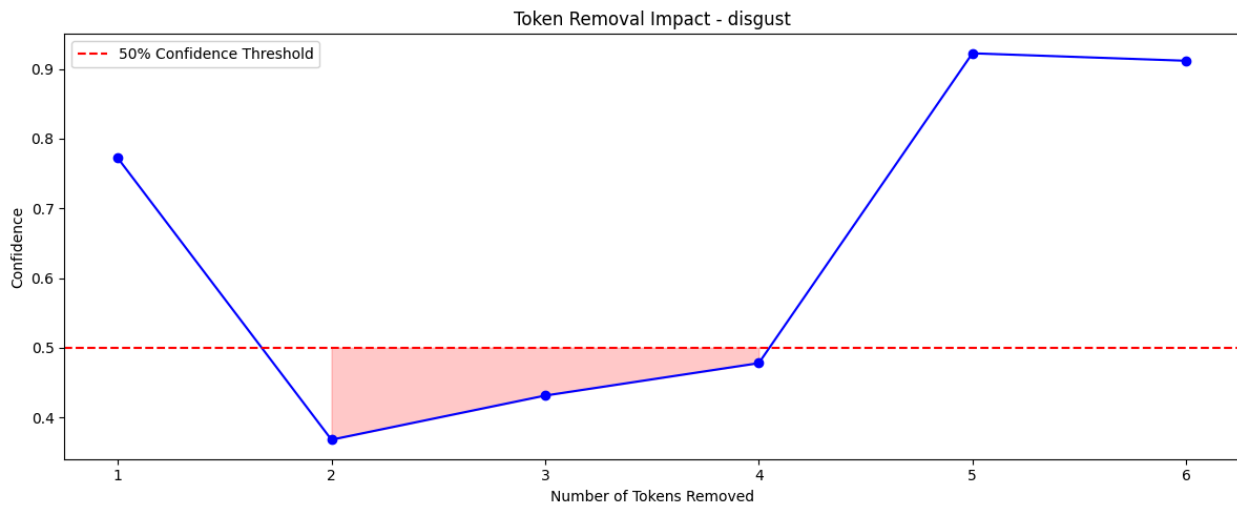
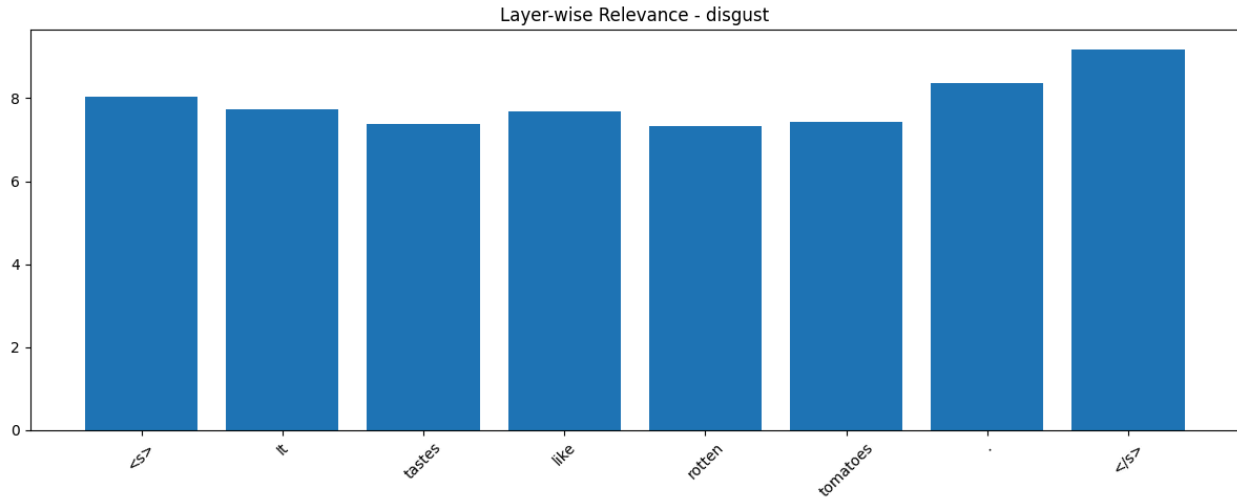




- "bad" having a strong positive influence aligns with my expectations. "smell" and "They" having negative influences are somewhat counterintuitive. While not directly indicating disgust, they contribute to the overall negative tone. I would need the entire context, at least "smell" and "bad" to properly classify this as "disgust". "They" and "smell" are not entirely irrelevant, but they are more direct indicators of disgust than "bad." The model still relies on self-attention of the words. In LRP, all tokens have moderate relevance scores (around 14-15), suggesting the model considers all parts of the sentence to contribute to the "disgust" classification, this seems to be a better analysis regarding my intuition as I do not consider "bad" to be enough to establish disgust and irrelevant tokens are not less emphasized. When the model removes ("They" and "smell"), its confidence in "disgust" remains barely above chance. It initially finds very little evidence of "disgust" in the sentence until the word "bad" is removed. The jump in confidence when "bad" is removed indicates that, despite being the last token removed, "bad" is the sole component for the model's "disgust" classification.

Predicted emotion: disgust

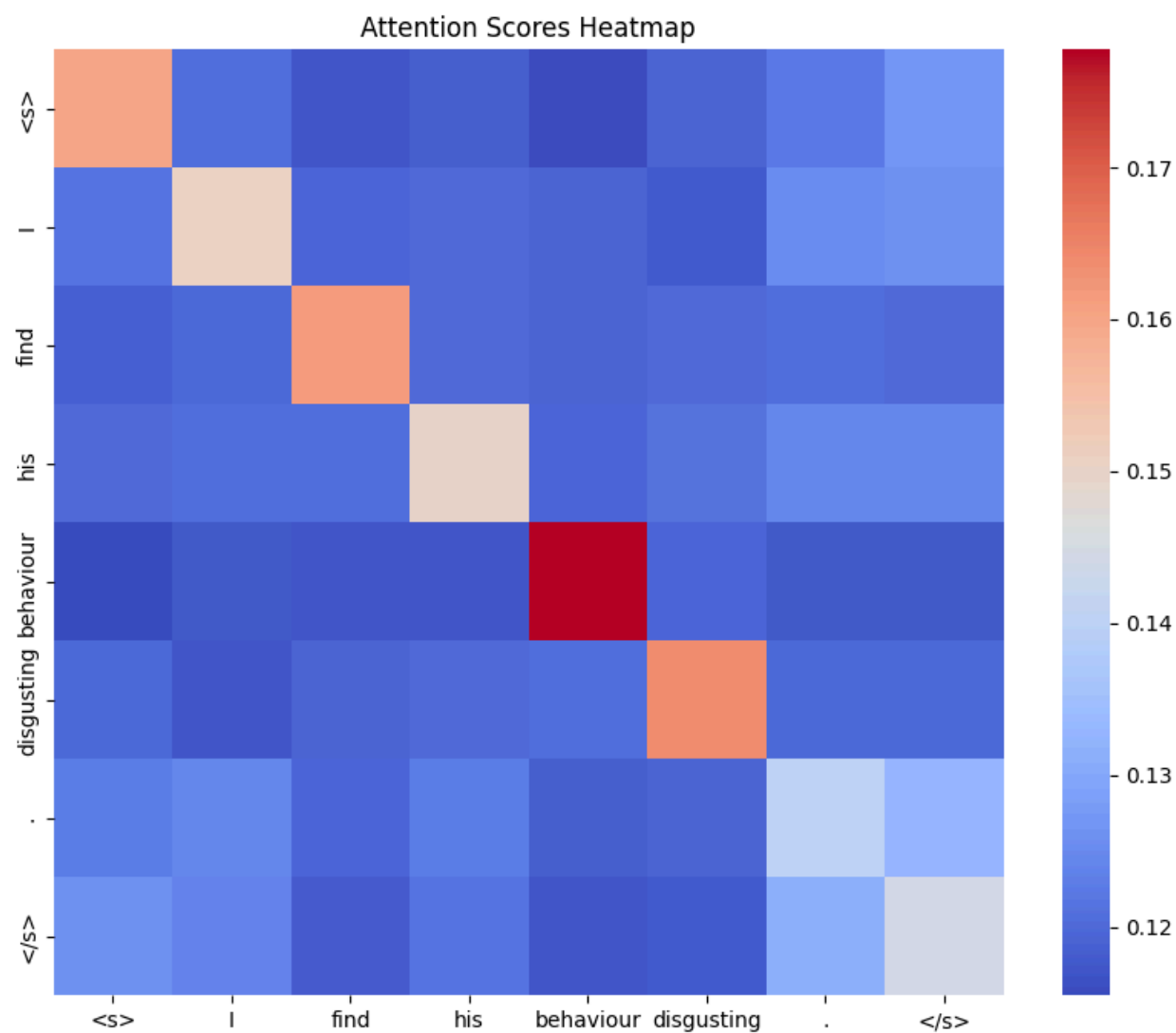
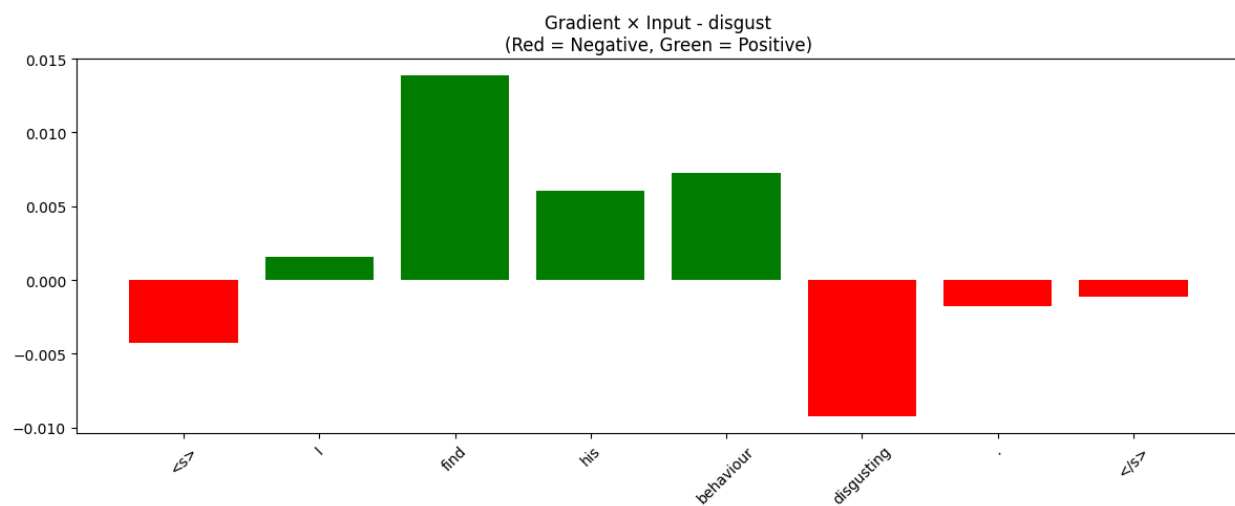


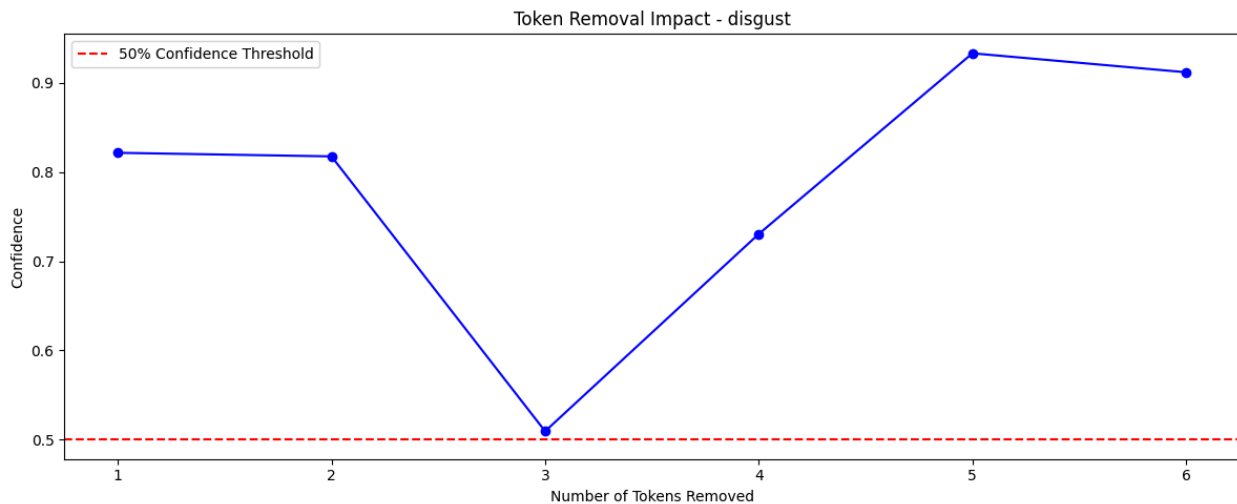
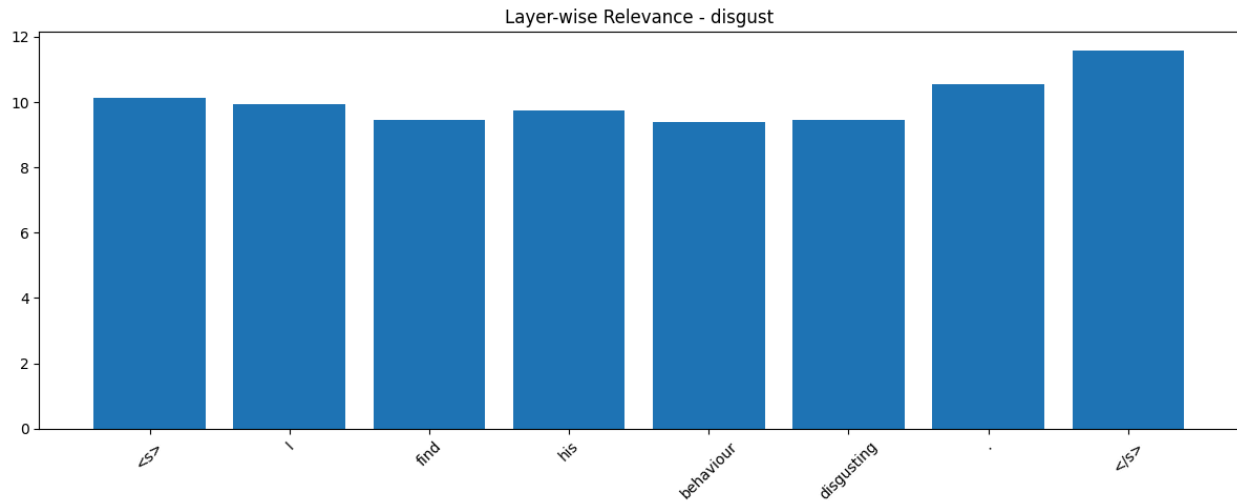


- The model incorrectly identifies "tomatoes" as a strong indicator of "disgust", because I do not find this stand-alone word to necessarily convey disgust but its interpretation of "rotten" is counterintuitive and doesn't align with my understanding. The context of "tastes like rotten" does not help the model at all to establish this classification and I find this not aligning with human perception. The model still relies on self-attention of the words so the model isn't fully grasping the interplay between words. In the "Layer-wise Relevance" chart, "tomatoes" and "rotten" are not emphasized as much as in "Gradient $\times$ Input." All tokens are considered relatively important, so it better values their overall context than the gradient input method. Confidence remains low until removing five tokens, then jumps dramatically, indicating "tomatoes" alone drives "disgust" recognition, revealing a potential overreliance on a single word.

Sentence 18: "I find his behaviour disgusting."

Predicted emotion: disgust





- The model correctly identifies "find," "his," and "behaviour" as contributing to "disgust," but its interpretation of "disgusting" is completely counterintuitive and doesn't align with my understanding. Less emotionally charged words also seem to be influencing the model's output such as "I" and ".". The model still relies on self-attention of the words so the model isn't fully grasping the interplay between words. The "Gradient $\times$ Input" method showed some impact from less emotionally charged tokens like "I" and ".". In the "Layer-wise Relevance" chart, these tokens have similar relevance to the emotionally charged ones, but offering these balanced views does not necessarily mean better classification. Starts confident in "disgust," then drops sharply when "his" is removed, showing its importance; confidence fluctuates with further removals, but jumps when only "disgusting" remains, showing its dominance as a "disgust" indicator.